Chemical Kinetics and Radioactivity

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Chemical Kinetics

It is the study of rate of reaction. its types, factors affecting rate and the mechanism followed by the reaction.

Rate of reaction (R.O. R)

For a chemical reaction, rate can be determined with respect to any reactant or any product. For a reaction, **the change in concentration (M)** of any reactant or product per unit time, **per unit stoichiometric coefficient**, is known as R.O.R.

$$\text{R.O.R} = \frac{\text{Change in conc. of any reactant or product}}{(\text{Time taken}) \, X (\text{ stoichimetric coefficient of same reactant or product}).}$$

Units :

$$\frac{M}{\text{time}} \text{ or } \frac{\text{Mole / L}}{\text{time}}$$

For gaseous reactions

We know that PV = nRT

$\Rightarrow$ P_{gas} = [gas] . R.T

For constant temperature

[concentration of gas] $\propto$ Partial pressure of gas

Means, for gaseous reaction, rate can also be written in terms of partial pressures as following –

$$\text{R.O.R} = \frac{\text{Change in partial pressire of gaseous component}}{\text{Time taken x stoichiometry coefficent of same component}}$$

Units : $\frac{\text{atm}}{\text{time}}$ or $\frac{\text{bar}}{\text{time}}$ or $\frac{\text{Pascal}}{\text{time}}$ etc.

Types of Rate of Reaction

(i) Average rate

The rate measured for a finite time interval (Δt).

i.e. Change in concentration $\Rightarrow$ finite

$\Rightarrow \Delta$(concentration)

Eg: For a reaction

x.A(g) $\longrightarrow$ y.B(g)

[Here, A is reactant with coefficient 'x' and B is product with coefficient 'y']

$$(\text{R.O.R})_{avg} = \left(\frac{-\Delta[A]}{\Delta t \, . \, x}\right) = \left(\frac{\Delta[B]}{\Delta t \, . \, y}\right) \Rightarrow \frac{M}{\text{time}}$$

Or $\left(\frac{-\Delta P_A}{\Delta t . x}\right) = \left(\frac{\Delta P_B}{\Delta t . y}\right) \Rightarrow \frac{P}{time}$

$$\boxed{(R.O.R.)_{avg.} = \left(\frac{\Delta P_A}{\Delta t.x}\right) = \left(\frac{\Delta P_B}{\Delta t.y}\right) \text{pressure / time}}$$

Note :

ΔA = value of change in concentration is negative hence rate will positive.

Negative sign shows decrease in concentration of A(reactant) w.r.t. time.

Positive sign increase charge in concentration of B (products) w.r.t time.

Example :

For gaseous reaction

	2A	⟶	3B
At t= 0	aM		–
At t = t	a–2x		3x

Rate for time (zero to t: 0 → t) (finite time):

$$\text{Rate} \Rightarrow (R.O.R)_{avg} = \left(\frac{-\Delta[A]}{\Delta t . \upsilon_A}\right)$$

or $\left(\frac{\Delta[B]}{\Delta t . \upsilon_B}\right)$

{υ → stoichiometric. coefficient }

$$= \frac{-[(a-2x)-(a)]}{(t-0).2} = \frac{[(3x-0)]}{(t-0).3}$$

$$\Rightarrow \frac{x}{t} = \frac{x}{t}$$

(ii) Instantaneous rate :

The rate measured at a particular instant of time.

i.e. time taken = very-very small (t→ 0)

= dt

⇒ Hence, change in concentration = very very small (d[ ])

i.e. for x.A(g) ⟶ y.B(g)

Points to remember

Hence for gaseous reactions rate of reaction (R.O.R.) can be expensed in two ways :

(a) w.r.t. concentration
(d) w.r.t. partical pressure

} Numerical valu will be different (units different)

Note : Standard rate of reaction is expressed in same of molarity or partial pressure.

Points to remember

Here negative sign indicates that amount or reactant is decreasing. But rate is positive.

Points to remember

Mode is avoided to distinguish between rate of reaction for reactant and product.

$$(R.O.R)_{inst.} = \left(\frac{-d[A]}{dt.x}\right) = \left(\frac{d[B]}{dt.y}\right) \Rightarrow \frac{M}{time}$$

Or $\left(\frac{-dP_A}{dt.x}\right) = \left(\frac{dP_B}{dt.y}\right)$

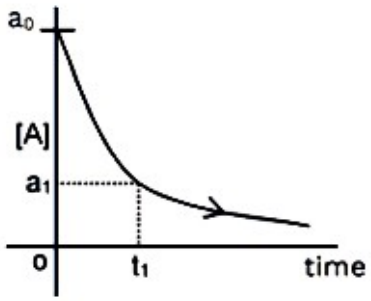
$\Rightarrow \frac{P}{time}$ { for gaseous reaction}

Points to remember

Concentration = $\frac{P}{(RT)}$

RT = 1 as T = 12.71 kelvin
Practically reaction not possible as above temperature.
Hence, RT > 1 Generally at [concentration] ≠ [Pressure]

Q1 **For A → B, Calculate Rate Of Reaction for time 0 → t_1**

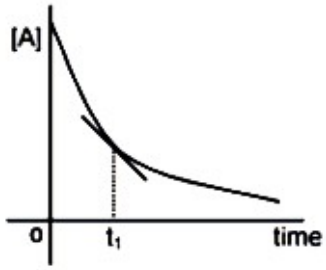


Sol: Since, 0 to t_1, is a finite value
i.e. Rate = $(ROR)_{avg.}$

$$= \left(\frac{-\Delta[A]}{\Delta t.V_A}\right) = \frac{-(a_1 - a_0)}{(t_1 - 0).1} = \frac{(a_0 - a_1)}{t_1}$$

Note: Since information given for A(reactant), rate of reaction $(R.O.R)_{avg}$ is positive as $a_0 > a_1$

Q2 **Calculate ROR at $t \to t_1$**



Sol: Rate = instantaneous i.e t_1.

$$= \left(-\frac{d[A]}{dt}_{.1}\right) = \frac{-d[A]}{dt}$$

$(R.O.R)_{instt}$ = – (slope of tangent at t_1) → (–)ve
= (+)

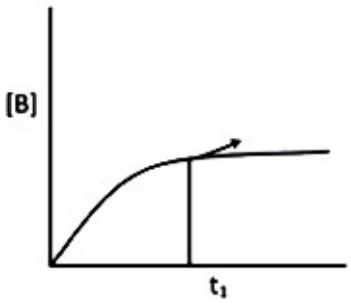
Note: for a curve: $\frac{dy}{dx}$ = slope

$(ROR)_{inst.}$ = slope
(structure of coefficient) for reactant.

Points to remember

Above relation is only & only applicable when rates are in terms of molarity, partial pressures or moles.

Q3 **For the reaction A → B , Calculate Rate of Reaction at time t= t_1**



Sol: $(ROR)_{inst.} = \left(+\frac{dB}{dt}\right)$

Slope is +ve. (products)
$(ROR)_{inst.}$ = slope of tangent at t_1

Rate of disappearance /decomposition (R.O.D)

Defined for reactants only as following –

$$(R.O.D)_{Reactant} = \frac{\text{Decrease in concentration}}{\text{Time taken}}$$

{decrease denoted by negative sign}

Example :

For xA + yB → Products

$$(R.O.D)_A = \left(-\frac{\Delta[A]}{\Delta t}\right) \text{ or } \left(-\frac{d[A]}{dt}\right)$$

$$(R.O.D)_B = \left(-\frac{\Delta[B]}{\Delta t}\right) \text{ or } \left(-\frac{d[B]}{dt}\right)$$

Note: stoichiometric coefficient is not used in above

Rate of disappearance (R.O.A)

Defined for products only as following –

$$(R.O.A)_{Prod.} = \frac{\text{Increase in concentration}}{\text{Time taken}}$$

i.e. Reactants → x.C + y.D

$$(ROA)_C = \left(\frac{\Delta[C]}{\Delta t}\right) \text{ or } \left(\frac{d[C]}{dt}\right)$$

{increase denoted by positive sign}

$$(ROA)_D = \left(+\frac{\Delta[D]}{\Delta t}\right) \text{ or } \left(+\frac{d[D]}{dt}\right)$$

Example :

For the reaction : $N_2 + 3H_2 \rightarrow 2\ NH_3$

In same time 1 mole of N_2 is consumed for every 3 mole of H_2 consumed simultaneously

Important note:

Rate of reaction is same for all reactants and products

Rate of appearance or rate of disappearance of products and reactant may or may not be the same.

For a reaction

$\therefore$ $(R.O.R) = \dfrac{\text{Change in concentration}}{\text{Time X(stoichiometric coefficient.)}}$

i.e. $R.O.R = \dfrac{(R.O.D)_{reactant}}{\upsilon_{reactant}} = \dfrac{(R.O.A)_{Product}}{\upsilon_{Product}}$

Example :

For a reaction : $2A + B \longrightarrow 4C + 3D$

$$R.O.R = \frac{(R.O.D)_A}{2} = \frac{(R.O.D)_B}{1} = \frac{(R.O.A)_C}{4} = \frac{(R.O.A)_D}{3}$$

Note : In above relation all rates must be of same type (average. or instantaneous.)

Q4 **For reaction : $N_2O_5 \rightarrow NO_2 + O_2$: in a certain time interval, the rate of decomposition of N_2O_5 is found 1.8×10^{-3} mol.L^{-1} min^{-1}, then what will be that rate of formation of O_2 during the same interval ?**

Sol: $2N_2O_5 \longrightarrow 4NO_2 + O_2$

$$\frac{(ROD)_{N_2O_5}}{2} = \frac{(ROA)_{O_2}}{1}$$

$$\frac{1.8 \times 10^{-3}}{2} = (ROA)_{O_2}$$

$$(ROA)_{O_2} = 9 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$$

Q5 **For a reaction ½ A → 2B, rate of disappearance of 'A' is related to the rate of appearance of 'B' by the expression :** **[AIEEE-2008]**

(A) $-\dfrac{d[A]}{dt} = \dfrac{1}{2}\dfrac{d[B]}{dt}$ **(B)** $-\dfrac{d[A]}{dt} = \dfrac{1}{4}\dfrac{d[B]}{dt}$

(C) $-\dfrac{d[A]}{dt} = \dfrac{d[B]}{dt}$ **(D)** $-\dfrac{d[A]}{dt} = 4\dfrac{d[B]}{dt}$

Sol: **(B)**

$$\frac{(ROD)_A}{\frac{1}{2}} = \frac{(ROA)_B}{2}$$

$$\frac{\left(\frac{-d[A]}{dt}\right)}{\frac{1}{2}} = \frac{\left(\frac{d[B]}{dt}\right)}{2} \Rightarrow \frac{-d[A]}{dt} = \frac{1}{4}\frac{d[B]}{dt}$$

Q6 **For reaction : $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$**
In an experiment, the concentration of NO increase by 1.08×10^{-2} mol/L in 3 seconds. Calculate,
(i) Rate of reaction
(ii) Rate of disappearance of ammonia
(iii) Rate of formation of steam

Sol: (i) $ROR = \left(\frac{\Delta[NO]}{\Delta t.4}\right) = \frac{1.08 \times 10^{-2}}{3 \times 4} = 9 \times 10^{-4}$ M/S

(ii) $\because ROR = \frac{(ROD)_{NH_3}}{4}$

$(ROD)_{NH_3} = 4 \times ROR$

$= 4 \times 9 \times 10^{-4}$

$\Rightarrow (ROD)_{NH_3} = 3.6 \times 10^{-3}$ M/S

(iii) $ROR = \frac{(ROA)_{H_2O}}{6}$

$\Rightarrow (ROA)_{H_2O} = 6 \times ROR$

$= 6 \times 9 \times 10^{-4}$

$(ROA)_{H_2O} = 5.4 \times 10^{-3}$ M / S

Q7 **The reaction $2NO(g) + Cl_2(g) \rightarrow 2NOCl(g)$ is carried out in a closed vessel. If partial pressure of NO is decreasing at rate of 180 torr/min then calculate rate of appearance of NOCl in torr/sec.**
(A) 180 **(B) 60** **(C) 30** **(D) None**

Sol: (D)

$\frac{(ROD)_{NO}}{2} = \frac{(ROA)_{NOCl}}{2}$

$(ROA)_{NOCl} = 180$ torr/min.

$= \frac{180}{60}$ torr/sec.

$= 3$ torr/sec.

Q8 **For reaction : $N_2 + 3H_2 \rightarrow 2\,NH_3$; if the rate of disappearance of N_2 at an instant is 5.6×10^{-4} gram / second then the rate of appearance of NH_3 in gram / second at that instant will be :**

(A) 1.12×10^{-3} **(B)** 2.0×10^{-5}

(C) 6.8×10^{-4} **(D)** 4.0×10^{-5}

Sol: **(C)**

$$\frac{(ROD)_{N_2}}{1} = \frac{(ROA)_{NH_3}}{2}$$

$$\frac{5.6 \times 10^{-4}}{28} = \frac{(ROA)_{NH_3}}{2}$$

$2 \times 2 \times 10^{-5}$ in mole/sec. $= (ROA)_{NH_3}$

$(ROA)_{NH_3} = 4 \times 10^{-5} \times 17$ in gm/sec.

$= 6.8 \times 10^{-4}$ gm/sec.

Lecture Note

First three factors majorly affect the rate. Hence, we will study them in detail in this chapter.

Q9 **In dilute aqueous H_2SO_4, the complex diaquodioxalateoferrate (II) is oxidized by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is :**

Sol: $\overset{+2}{[Fe(H_2O)_2(C_2O_4)_2]^{-2}} + \overset{+7}{M}nO_4^- \xrightarrow{[H^+]} Mn^{+2} + Fe^{+3} + \overset{+4}{C}O_2$

n.f. of $[Fe(H_2O_2)(C_2O_4)_2]^{-2} = 5$

n.f. of $MnO_4^- = 5$

Balanced reaction :

$[Fe(H_2O)_2(C_2O_4)_2]^{-2} + MnO_4^- + 8H^+ \rightarrow Mn^{+2} + Fe^{+3} + 4CO_2 + 6H_2O$

$$\frac{(ROD)_{MnO_4^-}}{1} = \frac{(ROD)_{H^+}}{8}$$

$$\frac{(ROD)_{H^+}}{(ROD)_{MnO_4^-}} = \frac{8}{1}$$

In short :

$$R.O.R = \frac{R.O.D}{\upsilon(\text{stoichiometric coefficient})} = \frac{R.O.A}{\upsilon(\text{stoichiometric coefficient})}$$

Factors affecting rate :

(1) Concentration of reactant
(2) Temperature
(3) Catalyst
(4) Partial pressure of gaseous reactant
(5) Surface area of solid reactant
(More Surface Area ⇒ More rate)
(6) Form of reactant used.
(7) Intensity of light (Only in photo chemical reaction) etc.

2. Rate law :

For a reaction at a constant temperature.

$a.A + b.B \longrightarrow$ Products

According to rate law :

$(Rate)_t \propto [A]_t^x [B]_t^y$

$\Rightarrow$ Rate $= k[A]^x [B]^y$ (Differential rate law expression)

Where,

$k \rightarrow$ Rate constant / velocity constant or specific rate

Also,

$x \rightarrow$ order w.r.t. A
$y \rightarrow$ order w.r.t. B
} Partial orders

& $\boxed{(x + y) \rightarrow \text{overall order of reaction } (n)}$

Example :

For $N_2 + 3H_2 \longrightarrow 2NH_3$

Rate $= k[N_2]^x [H_2]^y$

$\Rightarrow$ Since, 'x' and 'y' are unknown

$\Rightarrow$ incomplete rate law.

Points to remember

Since orders are determined only experimentally.
So rate law cannot be written without experimentally data.

i) Rate Constant (k) :

For a reaction, k is the measure of rate.

More value of k $\Rightarrow$ More rate

For a particular reaction k depends upon **temperature & Catalyst** only.

Units of k depend upon the overall order of reaction (n).

For, $A \longrightarrow$ Product

Rate $= k[A]^n$; (order of reaction = n)

$$\left(\frac{M}{time}\right) \equiv k(M)^n$$

$\Rightarrow$ Unit of k $\rightarrow M^{(1-n)}$ time^{-1}

i.e.

zero order	n = 0	k $\rightarrow$ M. time^{-1}
First order	n = 1	k $\rightarrow$ time^{-1}

ii) Order :

The power raised to the concentration in rate law expression is called partial order of that reactant. and

$$\boxed{\sum(\text{Partial order}) = \text{overall order of reaction}}$$

Since determined experimentally. So partial order of a reactant may be (+), (–), integer, fractional or even zero.

Calculation of order for different reaction:

Elementary / Simple reaction :

The reaction which takes place in one step only.
No intermediate is formed.

Example :

For elementary reaction :

a.A + b.B $\longrightarrow$ Product.

Experimentally :

Rate = k.$[A]^a$ $[B]^b$

i.e., Partial order = stoichiometric coefficient

where, a & b $\rightarrow$ Smallest integer

Points to remember

- But experimentally it is observed that overall order of reaction cannot be negative.
- For a particular reaction, on changing reaction conditions, order may be change.

Q10 **For the elementary reaction, 2A + 4B $\longrightarrow$ Product. Find the order of the reaction**

Sol: order $\neq$ (2 + 4)

but A + 2B $\longrightarrow$ product (simplest ratio)

order = (1 + 2) $\Rightarrow$ Order of the reaction = 3

2. Complex reaction :

Reaction which takes place in more than one steps.

At least one intermediate is formed.

Example :

$S_8 + 12O_2 \longrightarrow 8SO_3$

Mechanism:

Step 1: $S_8 + 8O_2 \longrightarrow 8SO_2$

Step 2: $8SO_2 + 4O_2 \longrightarrow 8SO_3$

Nett: $S_8 + 12O_2 \longrightarrow 8SO_3$

So, S_8 & O_2 ⇒ Reactants and SO_3 ⇒ Product and SO_2 ⇒ Intermediate.

Lecture Note

Each step of a complex reaction is an elementary reaction.

For complex reaction:

$a.A + b.B \longrightarrow$ Product

$\text{Rate} = k[A]^x[B]^y$

Where, $x \neq a$, $= a$ } all are possible according to experimental data; $y \neq b$, $= b$

Q11 **For reaction: A + 2B ⟶ C, the rate law is obtained as. Rate = k.[A] [B]², then**

(A) reaction must be elementary **(B) reaction must be complex**

(C) reaction may be elementary **(D) reaction may be complex**

Sol: (C, D)

Partial order = stoichiometric coefficient

Possible for elementary and complex reactions

Q12 **The reaction $2NO(g) + Cl_2(g) \rightarrow 2NOCl(g)$; the observed rate law is as following : Rate = $k[NO]^2[Cl_2]$ Then the reaction,**

(A) May be complex **(B) Must be elementary**

(C) Must be complex **(D) May be elementary**

Sol: **(A, D)**

since the order is same as that of the stoichiometric coefficient of the reactants.

Hence, the reaction can be complex or elementary

Points to remember

In question solving Until given, consider the reaction to be complex.

iii) Molecularity :

A theoretical concept defined only for elementary reaction.

The total number of reactant particles taking part in an elementary reaction to form product is known as **molecularity of the reaction.**

Molecularity cannot be negative zero or fractional means always will be a positive integer.

Elementary :

$a.A + b.B \longrightarrow$ Product

If a & b are smallest integer

$\Rightarrow$ Molecularity = (a + b)

Example :

$2A + B \rightarrow C$ } Molecularity = 2 + 1

2 mole + 1 mole } Whole number for stoichiometric coefficent

$A + \frac{1}{2}B \rightarrow \frac{1}{2}C$ molecularity = $1 + \frac{1}{2}$ (incorrect)

1 atom $\frac{1}{2}$ atom $\Rightarrow$ against Dalton's theory

Rate law

Elementary (Single step) | Complex (Multiple step)

Example:

$N_2O_4(s) \rightarrow 2NO_2$

The molecularity of above reaction = 1

Decomposition reaction gives molecularity one.

Example : $3A + B \rightarrow$ product

Mechanism I : $A + B \rightarrow C$.

molecularity = 2 (biomolecular)

Mechanism II : $2A + C \rightarrow D$.

molecularity = 3 (trimolecular)

Overall molecularity

= undefined for complex reaction.

Points to remember

- Molecularity = sum of state coefficient of all reactant's small integer for coefficient.
- Whole reaction must be balanced in terms of smallest integer.

Lecture Note

Experimentally, molecularity greater than 3 is not observed due is negligible probability of effective collision of more than 3 particles simultaneously. Possible values of molecularity = 1 or 2 or 3

For each step of a complex reaction, molecularity is defined but for overall complex reaction, it has no significance.

Points to remember

For a elementary reactions 1 simplest balanced order of reaction = molecularity = 1 or 2 or 3

i.e.,

if order = {zero or fractional or greater than 3} $\Rightarrow$ reaction must be complex.

Q13 **For the elementary reaction $2A + B_2 \rightarrow 2AB$, calculate how much the rate of reaction will change if the volume of the vessel is reduced to one third of its original volume?**

Sol: Rate $\propto [A]^2[B_2]^1$ [concentration = mol/litre]

$$\propto \left(\frac{1}{V}\right)^2\left(\frac{1}{V}\right)^1$$

$$(R_1)\ \text{Rate} \propto \left(\frac{1}{V}\right)^3$$

New rate and new volume = $\frac{V}{3}$

$$\text{New rate} \propto \left(\frac{1}{V/3}\right)^3$$

New rate (R_2) = 27 rate (R_1)

Rate Constants of appearance and disappearance:

Every type of rate can be written as following :

Rate = K [reactant] ^{order}

If $K = K_r$ (rate constant of reaction) $\Rightarrow$ Rate = (R.O.R)

If $K = K_d$ (rate constant of disappearance) $\Rightarrow$ Rate = (R.O.D)

If $K = K_a$ (rate constant of appearance) $\Rightarrow$ Rate = (R.O.A)

Eg. $N_2 + 3H_2 \longrightarrow 2NH_3$

$$ROR = k_r.[N_2]^x[H_2]^y$$

$$(ROD)_{N_2} = (k_d)_{N_2}[N_2]^x[H_2]^y$$

$$(ROD)_{H_2} = (k_d)_{H_2}[N_2]^x[H_2]^y$$

$$(ROA)_{NH_3} = (k_a)_{NH_3}[N_2]^x[H_2]^y$$

$$ROR = \frac{(ROD)_{N_2}}{1} = \frac{(ROD)_{H_2}}{3} = \frac{(ROA)_{NH_3}}{2}$$

For a reaction :

$$k_r = \frac{(k_d)_{Reactant}}{\upsilon_{Reactant}} = \frac{(k_a)_{Product}}{\upsilon_{Product}}$$

$\upsilon \rightarrow$ stoichiometric coefficient

Q14 **Dinitropentaoxide decomposes as follows :**

$$N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$$

If $\left(-\frac{d[N_2O_5]}{dt}\right) = k_1[N_2O_5]$; $\left(\frac{d[NO_2]}{dt}\right) = k_2[N_2O_5]$; $\left(\frac{d[O_2]}{dt}\right) = k_3[N_2O_5]$

What is the relation between k_1, k_2 and k_3 ?

Sol: $\because$ $\frac{(ROD)_{N_2O_5}}{1} = \frac{(ROA)_{NO_2}}{2} = \frac{(ROA)_{O_2}}{1/2}$

$$\frac{k_1[N_2O_5]}{1} = \frac{k_2[N_2O_5]}{2} = \frac{k_3[N_2O_5]}{1/2}$$

$$k_1 = \frac{k_2}{2} = 2k_3$$

Integrated rate law :

In its an expression between the concentration of reactant and time.

For A $\longrightarrow$ Products

At t = 0 (a) = A_0

At t = t (a – x) = A_t

Where,

a or A_0 = initial amount of reactant

(a - x) or A_t = Remaing amount of reactant at time t

consumed amount till time t = (x) or $(A_0 - A_t)$

} Molarity or Partial Pressure or moles

Degree of Completion (α) :

It is part of the reaction, which is completed at time t.

$$(\alpha)_t = \frac{\text{Consumed amount of reactant till time t}}{\text{Initial amount of reactant}}$$

i.e. $0 < \alpha \leq 1$

% Completion = $\alpha \times 100$

t_x or $t_{100x\%}$

Example :

$t_{\frac{1}{2}}$ or $t_{50\%}$, $t_{0.4}$, $t_{75\%}$, $t_{\frac{7}{8}}$ etc.

It is the time in which α-part of the initial amount of reactant is consumed.

$t_{\frac{1}{2}}$ = time in which half the amount of initial reactant is consumed.

$t_{\frac{7}{8}}$ = time in which $\frac{7}{8}$ of initial reactant is consumed.

$t_{75\%}$ = time in which 75% of initial reactant is consumed.

$\because$ $\boxed{\alpha = \frac{x}{a} = \frac{\text{amount of reactant consumed}}{\text{initial amount of reactant}}}$

$\Rightarrow$ $x = a\alpha$ or $A_0\alpha$

At any t_x

$A_t = a - x$ = amount of reactant left after time 't'

$= A_0 - A_0\alpha$

$\Rightarrow$ $\boxed{A_t = A_0(1-\alpha)}$

(1) Zero order kinetics :

For, $A \longrightarrow$ Product ; (order = 0)

Rate = $k.[A]^0$ (Must be complex)

$\Rightarrow$ $\boxed{\text{Rate} = k = \text{constant}}$

Now applying integrated rate law:

$$-\frac{d[A]}{dt} = k$$

$$-\int_{A_0}^{A_t} d[A] = k\int_{t=0}^{t=t} dt$$

$\Rightarrow$ $\boxed{A_0 - A_t = k.t}$

$(a) - (a - x) = k.t.$ or

$\Rightarrow$ $\boxed{x = k.t}$

For any t_α, put
$A_t = A_0(1 - \alpha)$ In integrated rate law

Points to remember

Means rate of a zero-order reaction remains constant throughout the reaction.
(Hint: Unitary Method can be applied)

Example :

To calculate $t_{\frac{1}{2}}$ or $t_{50\%}$ (Half-life)	$\alpha = \frac{1}{2}$ $\Rightarrow \quad A_t = A_0\left(1-\frac{1}{2}\right) = \frac{A_0}{2}$ Put $\quad A_0 - \frac{A_0}{2} = kt_{\frac{1}{2}}$ $\Rightarrow \quad \boxed{t_{\frac{1}{2}} = \frac{A_0}{2k}}$
To calculate $\mathbf{t_{\frac{3}{4}}}$ **or** $\mathbf{t_{75\%}}$	$\alpha = \frac{3}{4}$ $A_t = A_0\left(1-\frac{3}{4}\right) = \frac{A_0}{4}$ Put $\quad A_0 - \frac{A_0}{4} = k.t_{3/4}$ $\boxed{t_{\frac{3}{4}} = \frac{3A_0}{4.k}}$
$\mathbf{t_{100\%}}$ **(Completion time)**	i.e. $\quad \alpha = 1$ $A_t = A_0(1-1) = 0$ Put $\quad A_0 - 0 = k.t_{100\%}$ $\boxed{t_{100\%} = \frac{A_0}{k} = 2t_{50\%}}$

Note: Reactions having $0 \leq n < 1$ only have finite completion time.
i.e., If order $\geq 1 \Rightarrow$ Infinite completion time

Q15 **The rate constant for a zero-order reaction is 2×10^{-2} M.sec^{-1}. If the concentration of the reactant after 25 sec is 0.25M, calculate the initial concentration.**

Sol: Since, it is zero order reaction,
So, $A_0 - A_t = k.t$
$A_0 - 0.25 = 2 \times 10^{-2} \times 25$
$A_0 = 0.75$ M
$\Rightarrow$ Initial concentration (A_0)=0.75 M

Q16 **The half-life period of a zero-order reaction A → Product, is 1 hour when the initial concentration of the reactant 'A' is 2 M. How much time does it take for its concentration to come from 0.50 M to 0.25? [AIEEE-2010]**

(A) 0.25 h **(B) 1 h** **(C) 4 h** **(D) 0.5 h**

Sol: **(A)**

Case I: $A_0 - A_t = k.t.$

$2 - 1 = k \times 1$

$K = 1M/hour$

Case II: $A_0 - A_t = k.t.$

$0.5 - 0.25 = 1 \times t$

$t = 0.25$ hr.

or

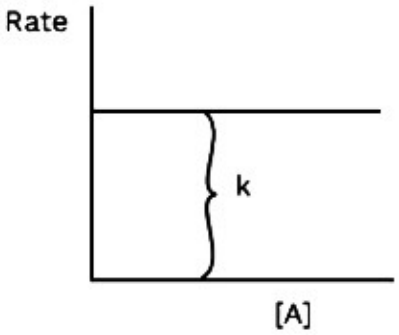
Unitary method

1M consume → 1 hr.

0.25 M consume → $\frac{1}{1} \times 0.25$ hr.

Graphs for zero order reactions :

(a) Rates v/s conc. for reaction



$Rate = k[A]^{\circ}$

(y) (x)

Comparing with equation of structure line

$y = mx + c$

$y = k.$

(b) [A] v/s time

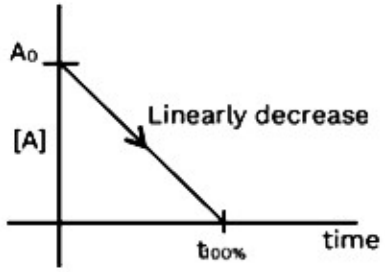
$\because \quad A_0 - A_t = k.t$

$y = -k.x + A_0$

$\because$ $y = mx + c$

i.e. slope (m) = $(-k) < 0$

Intercept (c) = $A_0 > 0$



(2) First order kinetics :

For A $\longrightarrow$ product

$\because$ order = 1

Rate = $k.[A]^1$

Rate $\propto$ [A]

$$-\frac{d[A]}{dt} = k.[A]$$

$$\int_{A_0}^{A_t} \frac{1}{[A]} d[A] = k \int_{t=0}^{t=t} dt$$

$\Rightarrow$ $\boxed{k.t. = \ln \frac{A_0}{A_t}}$

$\boxed{k.t. = 2.303 \log \frac{A_0}{A_t}}$

$\boxed{k.t. = \ln \frac{a}{a-x}}$

$\boxed{A_t = A_0.e^{-k.t.}}$

$t_{1/2}$ or $t_{50\%}$ (half-life):

$$\alpha = \frac{1}{2}$$

$$A_t = A_0\left(1-\frac{1}{2}\right) = \frac{A_0}{2}$$

Put $k.t._{\frac{1}{2}} = \ln\frac{A_0}{\frac{A_0}{2}}$

$\Rightarrow \boxed{t_{\frac{1}{2}} = \frac{\ln 2}{k} = \frac{0.693}{k}}$

For a 1st order reaction, every t_x is independent of initial concentration.

$t_\alpha = A_0(1-\alpha)$

$kt_\alpha = \ln\frac{A_0}{A_0(1-\alpha)}$

Let, $A_0 = 100 \left(t_{\frac{1}{2}} = 10\text{min}\right)$

10min↓ – 50%
A = 50
10min↓ – 50%
A = 25
10min↓ – 50%
A = 12.5
↓ and so on

Points to remember

Hence, we can say that the amount of reactant continuously decreases but is never zero.
Hence time of competition for first order reaction = infinity

Some important relation :

$\boxed{t_{75\%} \text{ or } t_{\frac{3}{4}} = 2.t_{\frac{1}{2}}}$

$\alpha = \frac{3}{4} \Rightarrow A_t = A_0\left(1-\frac{3}{4}\right)$

$k.t_{\frac{3}{4}} = \ln\frac{A_0}{A_{0/4}}$

$\boxed{t_{87.5\%} \text{ or } t_{\frac{7}{8}} = 3.t_{\frac{1}{2}}}$

Also, after 'x' half lives

$A_t = \frac{A_0}{2^x}$

Q17 **In a first order reaction, the concentration of the reactant decreases from 0.8 M to 0.4 M in 15 minutes. The time taken for the concentration to change from 0.1 M to 0.025 M is :** **[AIEEE-2004]**

(A) 30 min **(B) 60 min.** **(C) 7.5 min** **(D) 15 min**

Sol: **(A)**

For first order reaction

$$k.t. = \ln\frac{A_0}{A_t}$$

$$k \times 15 = \ln\frac{0.8}{0.4} \quad ...(i)$$

$$k \times t = \ln\frac{0.1}{0.025} \quad ...(ii)$$

On dividing equation (ii)/(i)

$$\frac{15}{t} = \frac{\ln 2}{2\ln 2} \Rightarrow t = 30 \text{ min}$$

or

$0.8\ M \longrightarrow 0.4\ M$

i.e. $t_{\frac{1}{2}} = 15$ min.

$0.1 \longrightarrow 0.025\ M$

i.e. $t_{\frac{3}{4}} = 2t_{\frac{1}{2}} = 30$ min.

Q18 **substance X decomposes following 1st order kinetics with a half life of 100 minutes. The fraction of initial concentration of X, which is reacted after 300 minutes will be :** **[AIEEE-2018]**

(A) 1/8 **(B) ½** **(C) ¼** **(D) 7/8**

Sol: Reacted $= (A_0 - A_t)$

$\Rightarrow$ Fraction of reacted $= \frac{(A_0 - A_t)}{A_0} = ?$

$$100 = \frac{\ln 2}{k} \Rightarrow k = \frac{\ln 2}{100}$$

$$\text{Now, } \left(\frac{A_t}{A_0}\right) = e^{-\frac{\ln 2}{100}\times 300} \Rightarrow e = 2^{-3} = \frac{1}{8}$$

$$\text{Fraction} = 1 - \frac{1}{8} = \frac{7}{8}$$

$$300 \text{ min} \Rightarrow 3t_{1/2} \Rightarrow A_t = \frac{A_0}{2^3} = \frac{A_0}{8}$$

Fraction = $\frac{A_0 - A_t}{A_0} = \frac{7}{8}$ $\Rightarrow$ fraction reacted after 300 minutes = 7/8

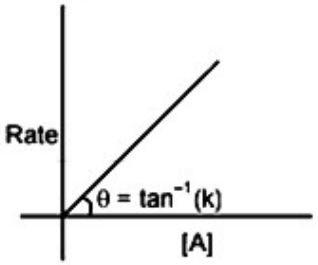
Graphs :

(1) Rate v/s [A]

For first order reaction : Rate = k $[A]^1$

On comparing it with equation of a straight line,

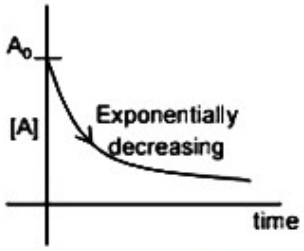
y = k.x slope=k intercept =0



(2) [A] v/s time

For first order reaction : $A_t = A_0 . e^{-k.t}$

$y \propto e^{-kx}$



Note: since time cannot be negative, so the graph lies in first quadrant

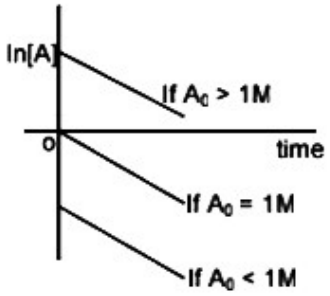
(3) ln [A] v/s time

$\because$ For first order reaction : k.t. = ln A_0 – ln A_t

y = – k.t. + ln A_0

$\because$ On comparing it with equation of a straight line, y = mx + C

Slope=-k and intercept =ln A_0



Note : All radioactive decay obey 1st order kinetics.

(3) Second order kinetics :

For, A $\longrightarrow$ Prod.

(order) n = 2

$$(2-1).\ k.t = \frac{1}{A_t^{(2-1)}} - \frac{1}{A_o^{(2-1)}}$$

$$\Rightarrow\ k.t. = \frac{1}{A_t} - \frac{1}{A_o}$$

At $t = t_{1/2}$ (half life)

$$A_t = A_o\left(1-\frac{1}{2}\right) \text{ put}$$

$$k.t_{1/2} = \frac{1}{\frac{A_o}{2}} - \frac{1}{A_o}$$

$$\boxed{t_{1/2} = \frac{1}{A_o.k}}$$

n^{th} order kinetics :

For, A $\longrightarrow$ Product

Rate = $k.\ [A]^n$

$$-\frac{d[A]}{dt} = k.[A]^n$$

$$-\int_{A_0}^{At}[A]^{-n}d[A] = k\int_{t=0}^{t=t} dt$$

$$-\left(\frac{[A]^{-n+1}}{-n+1}\right)_{A_0}^{A_t} = k.(t)_0^t$$

$$\Rightarrow\ \boxed{(n-1)\,k.t. = \frac{1}{A_t^{(n-1)}} - \frac{1}{A_0^{(n-1)}}\ (n \neq 1)}$$

t_x for n^{th} order :

$A_t = A_0(1-\alpha)$ put

$$(n-1).k.\ t_x = \frac{1}{A_0^{(n-1)}.\ (1-\alpha)^{(n-1)}} - \frac{1}{A_0^{(n-1)}}$$

$$\boxed{t_\alpha = \frac{1}{A_0^{(n-1)}}\left(\frac{1}{k.(n-1)}\left[\frac{1}{(1-\alpha)^{n-1}} - 1\right]\right)}$$

For a particular reaction:

$t_x \propto (A_0)^{1-n}$

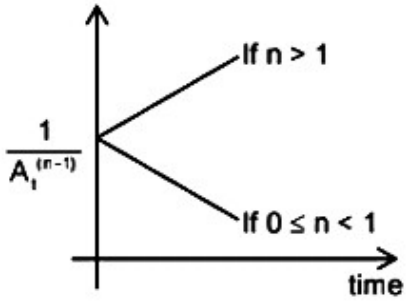
$\because$ (n – 1) k.t. $= \frac{1}{A_t^{(n-1)}} - \frac{1}{A_0^{(n-1)}}$

$y = (n-1)\,k.x + \frac{1}{A_0^{(n-1)}}$

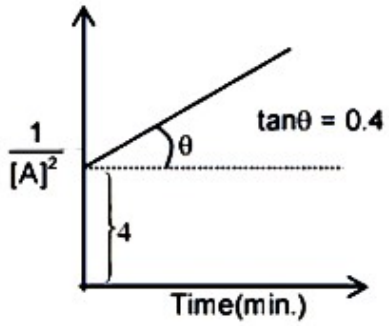
$\because$ $y = m\,x + c$

Intercept $= \frac{1}{A_0^{(n-1)}} > 0$

Slope = (n – 1). K



Q19 **For a reaction A → Product, a graph plotted between $\frac{1}{[A]^2}$ vs time is found to be linear with slope = 0.4 and y intercept equal to 4 as shown. The rate of disappearance of A at the initial stages will be :**



[A] is in terms of molarity.

(A) 0.025 M/s

(B) 4.16×10^{-4} M/s

(C) 1.6 M/s

(D) 0.8 M/s

Sol: **(B)**

On comparing with equation (n – 1) k.t. $= \frac{1}{A_t^{(n-1)}} - \frac{1}{A_0^{(n-1)}}$

$\frac{1}{[A]^{n-1}} = \frac{1}{[A]^2} \Rightarrow n - 1 = 2$

$\Rightarrow$ n = 3 i.e. it is 3rd order kinetics

Now,

$$(3-1).\ k.t = \frac{1}{A_t^{\,2}} - \frac{1}{A_0^{\,2}}$$

Intercept

$$\frac{1}{A_0^{\,2}} = 4$$

$\Rightarrow$ $A_0 = 0.5$ M

Also Slope $\Rightarrow (3-1).k = 0.4$

$k = 0.2\ M^{-2}\ min^{-1}$

Rate $= k.\ [A]^3$

$= 0.2 \times (0.5)^3\ M.\ min^{-1}$

$= \frac{0.2 \times (0.5)^3}{60} M.\ sec^{-1}$

$= 4.16 \times 10^{-4}\ M.sec^{-1}.$

Q20 **It was observed during the reaction of hydrogenation pressure of hydrogen decreases from 2 atm to 1.1 atm in 75 min. calculate rate of disappearance in molarity/sec. at 27°C.**

Sol:

$$(ROD)H_2 = \frac{-\Delta P_{H_2}}{\Delta t}$$

$$= -\frac{(1.1-2)}{75} atm/min.$$

$$= \frac{0.9}{75 \times 60} atm/sec.$$

$\because$ $P = [\ ]RT$

$$[\ ] = \frac{P}{RT} \Rightarrow \frac{\frac{0.9}{75 \times 60}}{0.0821 \times 300} M.\ sec^{-1}$$

$[\ ] = 8.1 \times 10^{-6}$ M/sec

Q21 **A elementary gaseous reaction $A(g) + 2B(g) \rightarrow C(g) + D(g)$**
Calculate the ratio of rate of reaction at time t and initial rate.
If, at $t = 0$ $P_A = 0.6$ $P_B = 0.8$
$t = t$ $P_C = 0.2$

Sol:

	$A_{(g)}$ +	$2B_{(g)}$ ⟶	$C_{(g)}$ +	$D_{(g)}$
$t = 0$	0.6	0.8	–	–
$t = t$	0.6–x	0.8–2x	x	x
$P_C =$	0.2 = x			

Rate $\propto (P_A)^1 (P_B)^2$

$$\frac{(\text{Rate})_t}{(\text{Rate})_0} = \left(\frac{P_{A_t}}{P_{A_0}}\right)^1 \left(\frac{P_{B_t}}{P_B}\right)^2$$

$$\frac{(\text{Rate})_t}{(\text{Rate})_0} = \frac{0.4}{0.6} \times \left(\frac{0.4}{0.8}\right)^2 \Rightarrow \frac{(\text{Rate})_t}{(\text{Rate})_0} = \frac{1}{6}$$

Note : Zero order: k.t. = $A_0 - A_t$

First order : k.t. = $\ln A_0 / A_t$

Second order : k.t. = $\frac{1}{A_t} - \frac{1}{A_0}$

n^{th} order : k.t. = $\frac{1}{(A_t)^{n-1}} - \frac{1}{(A_0)^{n-1}}$

} $k = k_d$ of reaction

k_d = rate constant of disappearance

$A \rightarrow$ product.

$\Rightarrow$ Above relations are valid when the coefficient of reactant is 1.

⇒ But if xA → products then above relations is not valid.

ROR = $k[A]^n$

$$\frac{1}{x}\left\{-\frac{d[A]}{dt}\right\} = k[A]^n$$

$[A]^{-n}\, d[A] = x.k.dt$

In short :

For nth order rection:

$$\boxed{(n-1)\, k.t = \frac{1}{A_t^{(n-1)}} - \frac{1}{A_o^{(n-1)}}}$$

Q22 **Decomposition of H_2O_2 follows a first order reaction. In fifty minutes $[H_2O_2]$ decreases from 0.5 to 0.125 M in one such decomposition. When the $[H_2O_2]$ reaches 0.05 M, the rate of formation of O_2 will be :**

[JEE MAIN-2016]

(A) 1.34×10^{-2} mol min^{-1}
(B) 6.93×10^{-2} mol min^{-1}
(C) 6.93×10^{-4} M min^{-1}
(D) 2.66 L min^{-1} at STP

Sol: **(C)**

$$k.t = \ln\frac{A_o}{A_t} \qquad \text{(first order reaction)}$$

$$k \times 50 = \ln\frac{0.5}{0.125}$$

$$k = \frac{\ln 2}{25}\,min^{-1} = \left(k_d\right)_{H_2O_2}$$

⇒ Rate = $k.[H_2O_2]^1$

$$\left(ROD\right)_{H_2O_2} = \frac{\ln 2}{25}(0.05)^1$$

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

$$\frac{(ROD)_{H_2O_2}}{1} = \frac{(ROA)_{O_2}}{\frac{1}{2}}$$

$$(ROA)_{O_2} = \frac{1}{2} \times \frac{0.693 \times 0.05}{25} M.min^{-1}$$

$$(ROA)_{O_2} = 6.93 \times 10^{-4} M.min^{-1}$$

Q23 **For a reaction : 2A(g) → B(g) + 3C(g), the rate constant of reaction is 5×10^{-4} $M.s^{-1}$. If initially 2 M of A is taken then what will be concentration of C after 5 minutes.**

(A) 0.3 M **(B) 0.9 M** **(C) 0.45 M** **(D) None**

Sol: **(C)**

	$2A \longrightarrow$	B +	3C ;	$5 \times 10^{-4} Ms^{-1} = k_r$
t = 0	2M	–	–	(from units of rate constant zero order)
t = t	2 – x	x/2	3x/2	

$\because k_r = \frac{(k_d)_A}{\upsilon_A}$

$\because k_d = 10^{-3} Ms^{-1}$

$\because$ Zero order $\Rightarrow x = k.t$

$\Rightarrow$ time = 5 × 60 = 300 sec.

$= 10^{-3} \times 300$

$x = 0.3$ M

$[C] = \frac{3x}{2}$

$= 0.45$ M

Concentration of C after 5 minutes = 0.45 M

Pseudo order reactions :

Applicable only for reactions having two or more reactants.

In such reactions, **if one of the reactants is taken in excess** w.r.t. other reactants then the **order of reaction** is changed & hence the reaction is called pseudo-order reaction.

Example :

$A + B \longrightarrow$ Prod. ; Rate = $k[A]^x[B]^y \Rightarrow$ order = $(x + y)$

$t = 0$ 1 1000

$t = t$ $(1 - a)$ $(1000 - a)$

$\because$ $a < 1$

$\Rightarrow$ $a << 1000$

$\Rightarrow$ $(1000 - a) \cong 1000$

i.e. [B] ~ Constant (rate is not dependent on concentration of B)

So Rate = $k_{net}[A]^x$

$\Rightarrow$ order = (x)

Where $k_{net} = k.[B]^y$

i.e. Pseudo x^{th} order reaction

Example :

$CH_3COOC_2H_5(aq.) + H_2O(l) \xrightarrow{H^+(aq)} CH_3COOH(aq) + C_2H_5OH(aq)$

Experimentally

Rate = k. [Ester] $[H_2O]$ $[H^+] \rightarrow$ Catalyst (Constant)

↓

Solvent

$\Rightarrow$ excess

$\Rightarrow$ Constant

Effectively

Rate = K_{net} [Ester]

i.e. pseudo first order reaction.

Where,

$k_{net} \propto [H^+]^1_{catalyst}$

$k_{net} = k[H_2O][H^+]$

$= K'[H^+]$

Order determination methods :

(1) Rate method :

Instantaneous rates are given at different concentrations of reactants.

Q24 **Consider a reaction aG + bH → Products. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is : [IIT JEE-2007]**

(A) 0 (B) 1 (C) 2 (D) 3

Sol: **(D)**

Let Rate = $k.[G]^x [H]^y$

Order of the reaction = (n+y)

$R_1=K[G_1]^x[H_1]^y$

$R_2=K[2G_1]^x[2H_1]^y$

$$\frac{R_1}{R_2} \Rightarrow \frac{1}{8} = \left(\frac{1}{2}\right)^x \times \left(\frac{1}{2}\right)^y$$

$\Rightarrow 1/8=(1/2)^{x+y} \Rightarrow (1/2)^3=(1/2)^{x+y}$

Over all order of the reaction $\Rightarrow (x + y) = 3$

Q25 **Initial rate data at 25ºC are listed in the table for reaction :**

$NH_4^+(aq) + NO_2^-(aq) \rightarrow N_2(g) + 2H_2O(l)$

Experiment	Initial $[NH_4^+]$	Initial $[NO_2^-]$	Initial rate (M/s)
1	0.24	0.1	7.2×10^{-6}
2	0.12	0.1	3.6×10^{-6}
3	0.12	0.15	5.4×10^{-6}

(a) What is the rate law ?

(b) What is the value of rate constant ?

(c) What is initial rate when initial concentration are $[NH_4^+]$ = 0.4 M and $[NO_2^-]$ = 0.5 M ?

Sol: Let, Rate = $k. [NH_4^+]^x [NO_2^-]^y$

From experiment I and II

$[NO_2-]$ → Same amount

Rate depends only on $[NH_4^+]$

Rate doubles as $[NH_4^+]$ doubles $\Rightarrow x = 1$

Similarly, **From experiment II and III**

Rate $\propto [NO_2^-]^y$

$$\frac{3.6 \times 10^{-6}}{5.4 \times 10^{-6}} = \left(\frac{0.1}{0.15}\right)^y$$

$$\Rightarrow \left(\frac{2}{3}\right) : \left(\frac{2}{3}\right)^y$$

$y = 1$

Rate = $k.[NH_4^+][NO_2^-]$

(b) solve for experiment I

Rate = $k.[NH_4^+][NO_2^-]$

$7.2 \times 10^{-6} = k(0.24)(0.1)$

$K = 3 \times 10^{-4}\ M^{-1}\ s^{-1}$

(c) Rate = $k.[NH_4^+][NO_2^-]$

rate = $3 \times 10^{-4} \times 0.4 \times 0.5$

$= 6 \times 10^{-5}$ M/s

(2) Hit and trial method :

Amounts will be given of different times.

$k.t. = A_o - A_t$ (for zero order)

$k.t = \ln\frac{A_o}{A_t}$ (for first order)

for given data in question solve if value of K is constant, then zero order reaction.

If not constant, solve for first order reaction and so on...

Q26 **The concentration of R in the reaction R → P was measured as a function of time and the following data is obtained :**

t (min.)	**0.0**	**0.05**	**0.12**	**0.18**
[R] (molar)	**1.0**	**0.75**	**0.40**	**0.10**

The order of the reaction is

Sol: Let the order of reaction n = 0

According to zero order reaction, $k.t = A_o - A_t$

at t= 0.05

$k \times 0.05 = 1 - 0.75 \Rightarrow K = 5$

at t = 0.12

$k \times 0.12 = 1 - 0.40 \Rightarrow K = 5$

in both cases value of K is same, hence it is zero orde reaction.

(3) Half life method :

in general, for a particular reaction.

any $\boxed{t_\alpha = A_0^{(1-n)}}$

Q27 **At certain temperature, the half-life period for the thermal decomposition of a gaseous substance depends on the initial partial pressure of the substance as follows :**

P (torr)	**500**	**250**
$t_{1/2}$ (min.)	**235**	**940**

The order of reaction is :

(A) 1 **(B) 2** **(C) 2.5** **(D) 3**

Sol: (D)

$t_{1/2} \propto (P_o)^{1-n}$

$$\frac{235}{940} = \left(\frac{500}{250}\right)^{1-n}$$

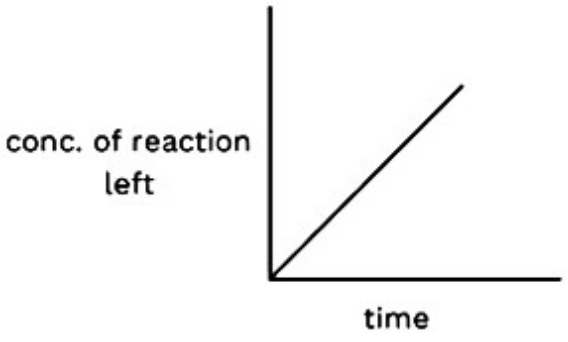
$2^{-2} = 2^{(1-n)}$

$\Rightarrow n = 3$

(4) Graphical method :

Graph will be given, and we have to identify the order

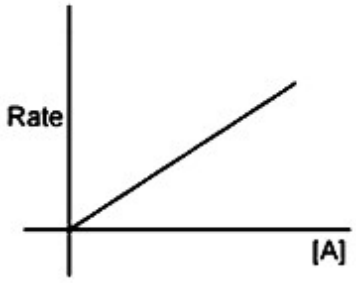
Example :



for zero order $(A_0 - At) = k.t.$

concentration of reactant left after time t = $(A_0 - A_t)$, satisfies zero order reaction

Example :



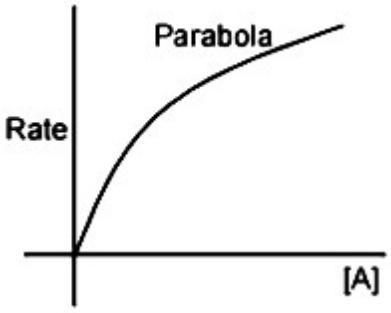
$\because$ from equation of a straight line: y = mx

Rate = m [A]

$\because$ Rate = k. $[A]^1$

i.e. order = 1

Example :



$\because$ $x \propto y^2$ (graph parabola)

$[A] \propto Rate^2$

$\Rightarrow$ Rate $\propto [A]^{1/2}$

i.e. $\frac{1}{2}$ order

Monitoring of first order reactions :

$$\boxed{k.t. = \ln \frac{A_0}{A_t}}$$

With respect is concentration can be solved in other parameter – pressure value or other properties for any first order reaction :

$$\boxed{k.t. = \ln\left(\frac{Q_\infty - Q_0}{Q_\infty - Q_t}\right)}$$

Where,

Q = any property (given in question)

k = k_d of reactant

(1) In terms of pressure :

Applicable for gaseous reactions taking place at constant temperature & volume.

Q28 **For reaction : A(g) → B(g) + C(g)**

Time	0	t
P_A	P_O	P_t

Calculate k

Sol:

$t = 0 \quad P_o$
$t = t \quad P_t$
$t = \infty \quad 0$

So, $k.t = \ln\left(\frac{0 - P_o}{0 - P_t}\right)$

$\Rightarrow \boxed{k = \frac{1}{t}\ln\frac{P_o}{P_t}}$

Q29 **For reaction : A(g) → B(g) + C(g)**

Time	O	t
P_{Total}	P_O	P_t

Calculate k

Sol:

$A(g) \rightarrow B(g) + C(g)$

$t = 0 \quad P_o \quad - \quad - \quad ; \quad P_o$
$t = t \quad - \quad - \quad - \quad ; \quad P_t$
$t = \infty \quad - \quad P_o \quad P_o \quad ; \quad 2P_o$

So, $k.t = \ln\frac{2P_o - P_o}{2P_o - P_t}$

$k = \frac{1}{t}\ln\frac{P_o}{2P_o - P_t}$

Q30 **For reaction : A(g) → B(g) + C(g)**

Time	t	∞
P_{Total}	P_t	P_∞

Calculate k

Sol:

	A(g) →	B(g) +	C(g)
t = 0	P_o	–	–
t = ∞	–	P_o	P_o

$P_x = 2P_o \Rightarrow P_o = \frac{P_\infty}{2}$

So, $k.t = \ln\frac{P_\infty - P_{\infty/2}}{P_\infty - P_t}$

$$k = \frac{1}{t}\ln\left(\frac{P_\infty}{2(P_\infty - P_t)}\right)$$

Q31 **For reaction : A(g) → 2B(g) + 3C(g)**

Time	O	t
P_{Total}	P_o	P_t

Calculate k

Sol:

	A(g) →	2B(g) +	3C(g)
t = 0	P_o	–	–
t = ∞	–	$2P_o$	$3P_o$

$\Rightarrow P_x = 5P_o$

So, $k.t = \ln\frac{5P_o - P_o}{5P_o - P_t}$

$$k = \frac{1}{t}\ln\frac{4P_o}{5P_o - P_t}$$

Q32 **For first order reaction : $2N_2O_5(g) \rightarrow 4NO(g) + 3O_2(g)$; taking place in a closed rigid vessel, after 80 hours from start of decomposition the total pressure developed is found to be 300 mm Hg. On complete decomposition, the total pressure is 350 mm Hg. Calculate the rate constant of the reaction. [Given : ln (5) = 1.6]**

Sol:

$$2N_2O_5(g) \rightarrow 4NO(g) + 3O_2(g)$$

$P_t = 300 \quad t = 0 \quad P_o \quad - \quad -$

$P_\infty = 350 \quad t = \infty \quad - \quad 2P_o \quad 3/2P_o$

$$\text{i.e. } P_\infty = 350 = 2P_o + \frac{3P_o}{2} \Rightarrow P_o = 100$$

$$\text{So, } k \times 80 = \ln\frac{350 - 100}{350 - 300}$$

$$k = \frac{\ln 5}{80} hr^{-1} \Rightarrow k_d$$

$$\because \quad kr = \frac{(k_d)_{N_2O_5}}{\upsilon_{N_2O_5}}$$

$$\Rightarrow \quad kr = \frac{\ln_{5/80}}{2} = \frac{\ln 5}{160} hr^{-1} = 0.01 \text{ hr}^{-1}$$

(2) Self decomposition of H_2O_2 :

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

The reaction obeys 1st order kinetics for which monitoring can be done in following two ways –

Q33 **For self decomposition of H_2O_2 following data are observed :**

Time	t	∞
Vol. of O_2 product at some constant T & P	V_t	V_∞

Calculate k

Sol: $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$

$\Rightarrow$ t = 0 - - 0 ml

t = t - - V_t

So, $k.t = \ln\left(\frac{V_\infty - 0}{V_\infty - V_t}\right)$

$k = \frac{1}{t}\ln\frac{V_\infty}{V_\infty - V_t}$

Q34 **For self decomposition of H_2O_2 following data are observed :**

Time	0	t
Vol. of $kMnO_4$ sol^n read for complete titration with H_2O_2	V_o	V_t

Calculate k

Sol: in titration applying law of equivalents:

Gm equivalent of $(aq.)_{KmnO_4} = (aq)_{H_2O_2}$

$\Rightarrow$ at t= infinity no H_2O_2 is left $V_\infty = 0$ i.e V_{KMnO4} = zero

So, $k.t = \ln\left(\frac{0 - V_o}{0 - V_t}\right)$

$\Rightarrow k = \frac{1}{t}\ln\frac{V_o}{V_t}$

(3) Hydrolysis of ester :

It is pseudo 1st order reaction when carried out in aqeous medium.

An acid-catalysed reaction means H^+ is used as catalyst

Ester(aq.) + $H_2O(l) \xrightarrow{H^+_{(aq)}}$ alcohol (aq.) + acid(aq.)

Q35 **For acid catalysed hydrolysis of $CH_3COOC_2H_5$;**

Time	0	t	∞
Vol. of a base solution required for complete neutralization	V_0	V_t	V_∞

Calculate k

Sol: $$k.t = \ln\frac{V_\infty - V_0}{V_\infty - V_t}$$

$$K = \frac{1}{t}\ln\frac{V_\infty - V_0}{V_\infty - V_t}$$

(4) Monitoring in terms of rotation :

Applicable for reactions consisting optically active compounds like glucose, sucrose etc. Each optically active compound has certain fixed value of its specific rotation which given as following –

$$\text{Specific rotation} = \frac{\text{Rotation}}{\text{Moles}}$$

Specific rotation is fixed for a solution.

For a solution containing n mole of Optically Active compound.

$$\text{Rotation} = \text{specific rotation} \times n$$

If solution containing more than one O.A. compounds.

$$\text{Total rotation} = \sum n_i.r_i$$

$n_i \rightarrow$ moles

$r_i \rightarrow$ sp. rotation

Q36 **For hydrolysis of sucrose in aqueous medium ;**

Time	0	t	∞
Total rotation	r_0	r_t	r_∞

Calculate k

Sol: $\Rightarrow k.t = \ln\left(\frac{r_\infty - r_o}{r_\infty - r_t}\right)$

$C_{12}H_{22}O_{11}(aq.) + H_2O(l) \longrightarrow C_6H_{12}O_6(aq.)\ C_6H_{12}O_6(aq)$

	Sucrose (r_s)	Glucose(r_G)	Fructose(r)
t = 0	a mole	—	—
t = t	a – x	x	x
t = ∞	—	a	a

$\because\ k.t = \ln\frac{a}{a-x}$ ..... (1)

$\because\ r_{total} = \sum n_i.r_i$

Where ,

r_S, r_G, r_F are specific rotation of sucrose, glucose, fructose respectively

t = 0

$r_o = a.\ r_S + o.\ r_G + o.\ r_F$ ..... (2)

t = t

$r_t = (a - x).\ r_S + (x).\ r_G + (x).\ r_F$ ..... (3)

$r_\infty = 0 + (a)\ r_G + (a)\ r_F$ ..... (4)

(4) – (2)

$(r_\infty - r_o) = a(r_G + r_F - r_S)$

(4) – (3)

$(r_\infty - r_t) = (a - x)\ (r_G + r_F - r_S)$

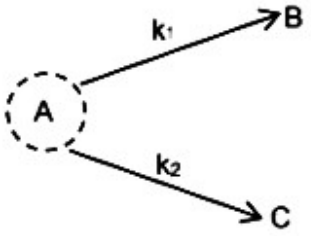
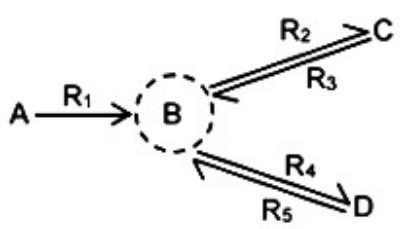
Divide,

$\frac{r_\infty - r_o}{r_\infty - r_t} = \left(\frac{a}{a-x}\right)$ Put in (1)

$\Rightarrow k.t. = \ln\frac{r_\infty - r_o}{r_\infty - r_t} \quad \Rightarrow K = \frac{1}{t}\ln\frac{r_\infty - r_o}{r_\infty - r_t}$

Combined reactions :

- Simultaneously running multiple reactions in which, some substances are common (at least one).
- For the common substance net R.O.D or net R.O.A is written as following –

Eg. 	Rate $\Rightarrow R_1 = \left(-\frac{d[A]}{dt}\right) = \left(\frac{d[B]}{dt}\right)$ $R_2 = \left(-\frac{d[A]}{dt}\right) = \left(\frac{d[C]}{dt}\right)$ Now, $\left(\frac{d[B]}{dt}\right)_{net} = R_1$ $\left(\frac{d[C]}{dt}\right)_{net} = R_2$ and $\left(-\frac{d[A]}{dt}\right)_{net} = R_1 + R_2$
Eg. $A \underset{R_2}{\overset{R_1}{\rightleftharpoons}} B$	$R_1 = \left(-\frac{d[A]}{dt}\right) = \left(\frac{d(B)}{dt}\right)$ $R_2 = \left(-\frac{d[B]}{dt}\right) = \left(\frac{d(A)}{dt}\right)$ Now, $\left(-\frac{d(A)}{dt}\right)_{net} = R_1 - R_2$ $\left(\frac{d(A)}{dt}\right)_{net} = R_2 - R_1$ $\left(\frac{d[B]}{dt}\right)_{net} = \mathbf{R_1 - R_2}$
Eg. 	$\left(\frac{d[B]}{dt}\right)_{net} = R_1 - R_2 + R_3 - R_4 + R_5$

Eg. $2A \underset{R_2}{\overset{R_1}{\rightleftharpoons}} 3B$	$R_1 = \frac{1}{2}\left(-\frac{d[A]}{dt}\right) = \frac{1}{3}\left(\frac{d(B)}{dt}\right)$ $R_2 = \frac{1}{3}\left(-\frac{d[B]}{dt}\right) = \frac{1}{2}\left(\frac{d(A)}{dt}\right)$ Now, $\left(\frac{d(A)}{dt}\right)_{net} = 2R_1 - 2R_2$ $\left(-\frac{d(B)}{dt}\right)_{net} = 3R_2 - 3R_1$

Q37 **For an elementary chemical reaction, $A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2A$, the expression for $\frac{d[A]}{dt}$ is : [JEE Main 2019-Jan.]**

(A) $k_1[A_2] + k_{-1}[A]^2$ **(B)** $2k_1[A_2] - 2k_{-1}[A]^2$

(C) $2k_1[A_2] - k_{-1}[A]^2$ **(D)** $k_1[A_2] - k_{-1}[A]^2$

Sol: **(B)**

$$R_1 = \frac{1}{2}\left(\frac{d[A]}{dt}\right) = k_1[A_2]^1$$

$$R_2 = \frac{1}{2}\left(-\frac{d[A]}{dt}\right) = k_2[A]^2$$

$$\text{Now, } \left(\frac{d[A]}{dt}\right) = 2k_1[A_2]^1 - 2k_2[A]^2$$

Q38 **For an elementary reaction scheme $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, if the rate of formation of B is set to be zero then the concentration of B is given by :**

(A) $k_1k_2[A]$ **(B)** $(k_1 + k_2)[A]$ **(C)** $(k_1 - k_2)[A]$ **(D)** $\left(\frac{k_1}{k_2}\right)[A]$

Sol: **(D)**

$$R_1 = \left(\frac{d(B)}{dt}\right) = k_1[A]^1$$

$$R_2 = \left(-\frac{d(B)}{dt}\right) = k_2[B]^1$$

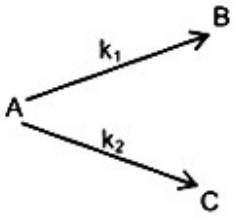
$$\left(\frac{d(B)}{dt}\right)_{net} = k_1[A] - k_2[B]$$

$$\Rightarrow [B] = \frac{k_1}{k_2}[A]$$

Parallel 1st order reactions :

Simultaneously running multiple 1st order reaction having same reactants but different products.

Example :



At t = 0	(a) = A_o
	$- x_1$
	$- x_2$
at t = t	(a − x) = A_t

$$\left(-\frac{d[A]}{dt}\right) = k.[A]$$

Where,

$x = (x_1 + x_2)$

Total consumed amount till time.

$\because$ x_1, x_2 are consumed amounts in same time period in reaction 1 and 2 respectively.

Both reactions going simultaneously, hence x_1 and x_2 will depend on speed of reaction.

So, $\boxed{x_1 : x_2 = k_1 : k_2}$

Where,

all $k \rightarrow k_d$

So, $x_1 = \left(\frac{k_1}{k_1 + k_2}\right).x$

$x_2 = \left(\frac{k_2}{k_1 + k_2}\right).x$

where,

$x = (A_o - A_t)$

now,

$$\left(-\frac{d[A]}{dt}\right)_{net} = R_1 + R_2$$

$= k_1[A]^1 + k_2[A]^1$ ($\because$ Given 1st order)

$$-\frac{d[A]}{dt} = k_{net}[A]^1$$

$$\int_{A_o}^{A_t} d[A] = k_{net}\int_{t=0}^{t=t} dt$$

$$\Rightarrow \boxed{k_{net}t = \ln\frac{A_o}{A_t}}$$

Where,

$K_{net} = k_1 + k_2$

Net rate constant for disappearance of A

$\because$ For 1st order, $t_{\frac{1}{2}} = \frac{\ln 2}{k}$

So, $$\boxed{\frac{\ln 2}{\left(t_{\frac{1}{2}}\right)_{net}} = \frac{\ln 2}{\left(t_{\frac{1}{2}}\right)_1} + \frac{\ln 2}{\left(t_{\frac{1}{2}}\right)_2}}$$

Also, $$\left(t_{\frac{1}{2}}\right)_{net} = \frac{\ln 2}{k_{net}}$$

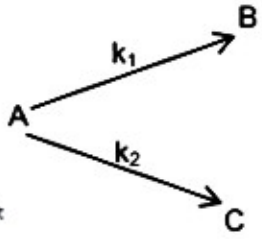
$$\Rightarrow \boxed{\frac{1}{\left(t_{\frac{1}{2}}\right)_{net}} = \frac{1}{\left(t_{\frac{1}{2}}\right)_1} + \frac{1}{\left(t_{\frac{1}{2}}\right)_2}}$$

% distribution of A

Into 1st reaction $A \rightarrow B = \frac{x_1}{x} \times 100$

$$= \left(\frac{k_1}{k_1 + k_2}\right) \times 100$$

Into 2nd reaction $A \rightarrow C = \left(\frac{k_2}{k_1 + k_2}\right) \times 100$



$A_o - x_1 = x_2 = A_{2t}$

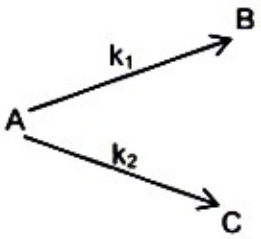
$(A_o - x) = A_t$

$$\boxed{k_{net}.t = \ln\frac{A_o}{A_t}}$$

$$x_1 = \left(\frac{k_1}{k_1 + k_2}\right).x$$

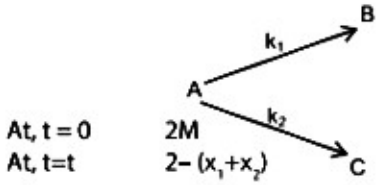
$$x_2 = \left(\frac{k_2}{k_1 + k_2}\right).x$$

Q39 **For,**



If $k_1=3k_2$, and reaction, starts with 2M of A. Then calculate [B] and [C] when concentration of [A]=0.4M

Sol:



At, $t=t - x_2$

$(2 - x) = A_t = 0.4$ M

$\Rightarrow$ $x = 1.6$ M = amount of A consumed in time t

If, $k_1 = 3k_2$ & the reaction is started with 2M of A. consumed [B] & [C] when [A] = 0.4 M.

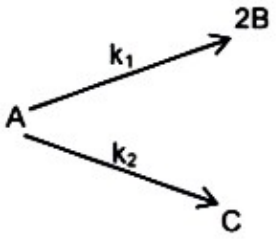
$$x_1 = \left(\frac{k_1}{k_1 + k_2}\right).x = \text{amount of A consumed to form B}$$

$$= \frac{3}{4} \times 1.6 = 1.2\,M$$

$x_2 = (1.6) - 1.2 = 0.4$ M = amount of A consumed to form C

Now, [B] = x_1 = 1.2 M, [C] = x_2 = 0.4 M

Q40



Started with 0.1 M of A, if after t time.

[A] = 0.05 M = A_t

[B] = 0.01 M then $\frac{k_1}{k_2}$ = ?

Sol: $\Rightarrow$ $x = (A_o - A_t) = 0.05$ M

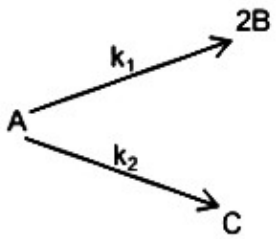
$$x_1 = 2.\left(\frac{k_1}{k_1 + k_2}\right).x$$

$\because$ [B] = $2x_1$

$$x_1 = \frac{0.01}{2}$$

$$0.01 = 2.\left(\frac{k_1}{k_1 + k_2}\right) \times 0.05$$

$$\frac{k_1}{k_2} = \frac{1}{9}$$



Transition state theory :

It gives explanation about the mechanism of **a bimolecular gaseous elementary reaction**.

Example $A_{(g)} + B_{(g)} \xrightarrow{\text{elementary}} 2\,AB_{(g)}$

$(A - A)\ (B - B) \qquad\qquad (A - B)$

Chemical reaction is a rearrangement of bonds.

Method I:

Step 1: Bonds break → 2A atoms and 2 B atoms are formed

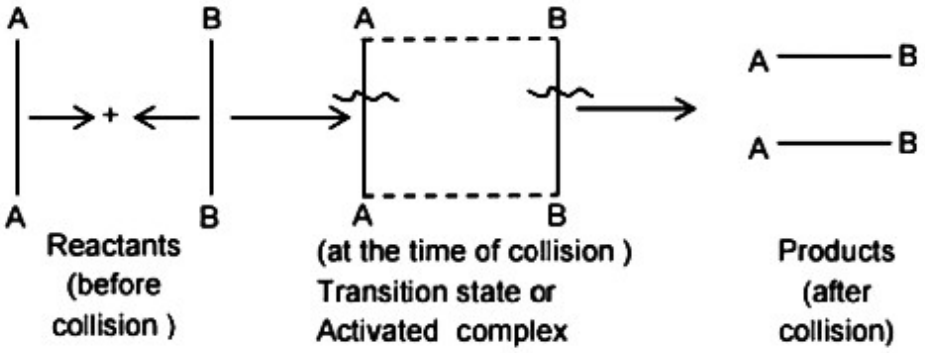
Step 2: New bonds between A and B atoms are formed.

Since, the above reaction is a multistep reaction. The above mechanism will not hold true for an elementary reaction.

Method II :

The only possible mechanism for above elementary reaction is as following

To form product A_2 and B_2 molecule should collide with each other



At the time of collision, new bond are formed between A---B , simultaneously old bonds break.

The stage is called as **transition state or activated complex**.

Means in elementary reaction, reactants always convert into products through transition state.

Transition state :

Re-arrangement of bonds takes place in this state.

This is the least stable state in an elementary reaction hence its Potential Energy is greater than that of both reactants and products.

Threshold energy of reaction (E_{Th}) :

The minimum total energy that must be possessed by reactant particles to convert into products.

Since to convert into products., reactant particles must reach to Transition State means

Potential Energy of Transition State is equal to **the threshold energy of the reaction.**

i.e. $\boxed{E_{th} = \text{ Potential Energy of Transition State}}$

Activation energy of reaction (Ea) :

The **minimum Kinetic Energy** that must be possessed by reactant particles to reach upto Transition State.

$\because$ To reach T.S.

$(TE)_{Reactant} \geq E_{Th}$

$(KE + PE)_{Reactant} \geq E_{Th}$

$(KE)_{Reactant} \geq (E_{Th} - E_R)$

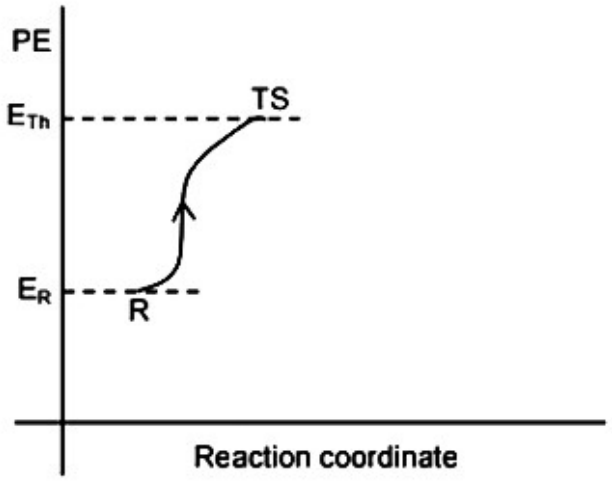
$$\boxed{\text{Min. KE} \Rightarrow (E_{Th} - E_e) = E_a}$$

Where,

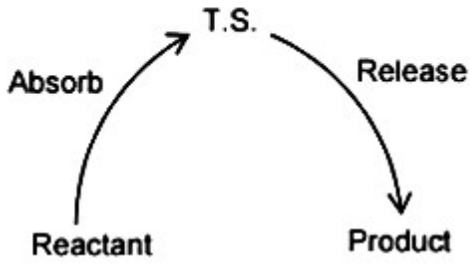
$E_R \rightarrow$ PE of reactant

For a particular reaction P.E of all reactant particles are equal but their K.E are not equal. So only those reactant particles will convert into products for which $KE \geq (E_a)_{Reaction}$

For an elementary reaction :

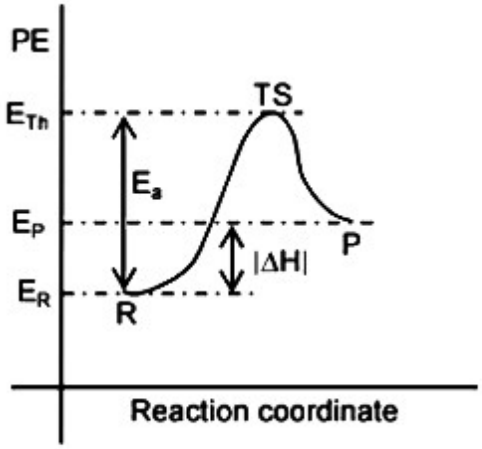


The P.E of product will be lesser than that of T.S, but may be greater or lesser than that of E_R. So, there are two types of reactions as following –



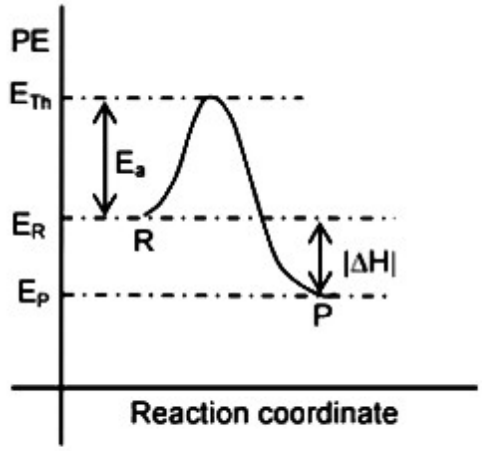
Endothermic ($\Delta H > 0$)

$\Rightarrow E_P > E_R$



Endothermic ($\Delta H < 0$)

$\Rightarrow E_R > E_P$



Note : For a reaction

$\Delta H = E_P - E_R$

$E_R \rightarrow$ PE of reactant

$E_P \rightarrow$ PE of product

Note : For endothermic reaction,

$E_a > |\Delta H|$ always

But for exothermic reaction, E_a may be greater or lesser or even equal to $|\Delta H|$.

Arrhenius equation : For a reaction

$$\boxed{k = A.e^{-\frac{E_a}{RT}}}$$

Where,

T = kelvin temperature

R = gas constant (8.314 J/mole.k or 2 cal/mol.k)

E_a = activation energy of reaction (J/mol or cal/mol)

A = Arrhenius factor/per-exponential factor

K = Rate constant factor/per-exponential factor

Pre-exponential factor (A) :

It is the measure of rate of collisions among reactant particles irrespective of their energy.

i.e. $\boxed{A \propto \text{Rate of total collisions}}$

$e^{-\frac{E_a}{RT}}$ →It is equal to the fraction of reactant particles which are in activated state at T.Kelvin.

i.e. $\boxed{\left(e^{-\frac{E_a}{RT}}\right) = \frac{\text{Particles having KE} \geq E_a}{\text{Total number of reactant particles}}}$

$$\% \text{ in activated state} = \left(e^{-\frac{E_a}{RT}}\right) \times 100$$

Also, $\ln k = \left(A.e^{-\frac{E_a}{RT}}\right)$

$$\Rightarrow \ln k = \ln A - \frac{E_a}{RT}$$

& $\boxed{\log k = \log A - \frac{E_a}{2.303RT}}$

For ln k v/s 1/T graph :

$$\ln k = \ln A - \frac{E_a}{R}.\frac{1}{1}$$

$$y = C + m\,x$$

$$\text{So, Slope(m)} = \left(-\frac{E_a}{R}\right) < 0$$

$$C = \ln(A) > 0$$

Q41 **For a first order reaction A → P, the temperature (T) dependent rate constant (k) was found to follow the equation :**

$$\log k = -(2000)\,1/T + 6.0$$

The pre-exponential factor A and the activation energy E_a respectively are

(A) $1.0 \times 10^6\ s^{-1}$ and 9.2 kJ mol^{-1}

(B) $6.0\ s^{-1}$ and 16.6 kJ mol^{-1}

(C) $1.0 \times 10^6\ s^{-1}$ and 16.6 kJ mol^{-1}

(D) $1.0 \times 10^6\ s^{-1}$ and 38.3 kJ mol^{-1}

Sol: **(D)**

On comparing with equation,

$$\log k = \frac{-E_a}{2.303RT} + \log A$$

So, $\log A = 6 \Rightarrow A = 10^6\ s^{-1}$

Also, $\frac{-E_a}{2.303R} = -2000$

$\Rightarrow E_a = 38.3$ kJ/mol

Q42 **Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300K, then $\ln(k_2/k_1)$ is equal to :**

(A) 12 **(B) 6** **(C) 4** **(D) 8**

Sol: **(C)**

$E_1 = E_2 + (10 \times 10^3\ J)$

$A_1 = A_2$

$$k_1 = A_1 . e^{-\frac{E_1}{RT}}$$

$$k_2 = A_2 . e^{-\frac{E_2}{RT}}$$

$$\frac{k_2}{k_1} = e^{-\frac{(E_2 - E_1)}{RT}}$$

$$\ln \frac{k_2}{k_1} = \frac{(E_2 - E_1)}{RT}$$

$$\Rightarrow \frac{10 \times 10^3}{\frac{25}{3} \times 300} = 4$$

Effect of temperature :

For a particular reaction,

$E_a \rightarrow$ Independent of temp.

$A \rightarrow$ Negligible change with temp. (~ constant)

$\because$ according to Arrhenius equation

$$k = A . e^{-\frac{E_a}{RT}}$$

So, upon changing temperature, the value of k will change.

Let, $T = T_1$ then $\ln k_1 = \ln A - \frac{E_a}{RT_1}$ ... (1)

And $T = T_2$ then $\ln k_2 = \ln A - \frac{E_a}{RT_2}$ ... (2)

On subtracting equations (1) from (2)

So, $\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

Or, $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

Let temperature increases $\Rightarrow T_2 > T_1$

$$\left(\frac{1}{T_1} - \frac{1}{T_2}\right) > 0$$

Also, $E_a > 0$, $R > 0$

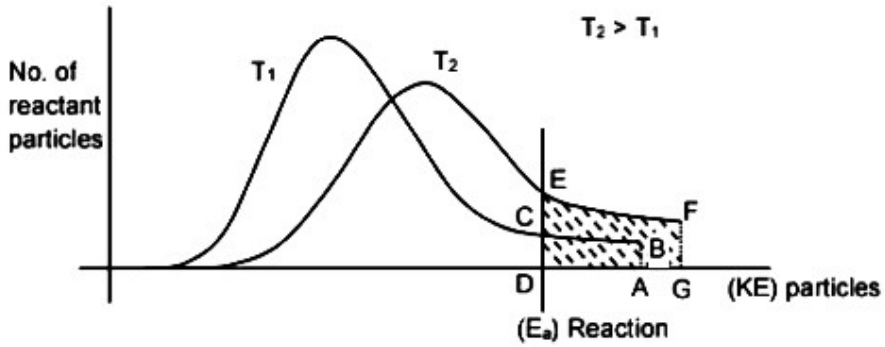
$\Rightarrow \ln \frac{k_2}{k_1} > 0 \Rightarrow$ Hence, $r_2 > r_1$

Points to remember

on increasing temp., k increases $\Rightarrow$ Rate increases & vice-versa

This is because, on increasing temperature, KE of all reactant particles increase so the number of particles having KE $\geq E_a$. will increase & hence k↑.

From Maxwell's distribution :

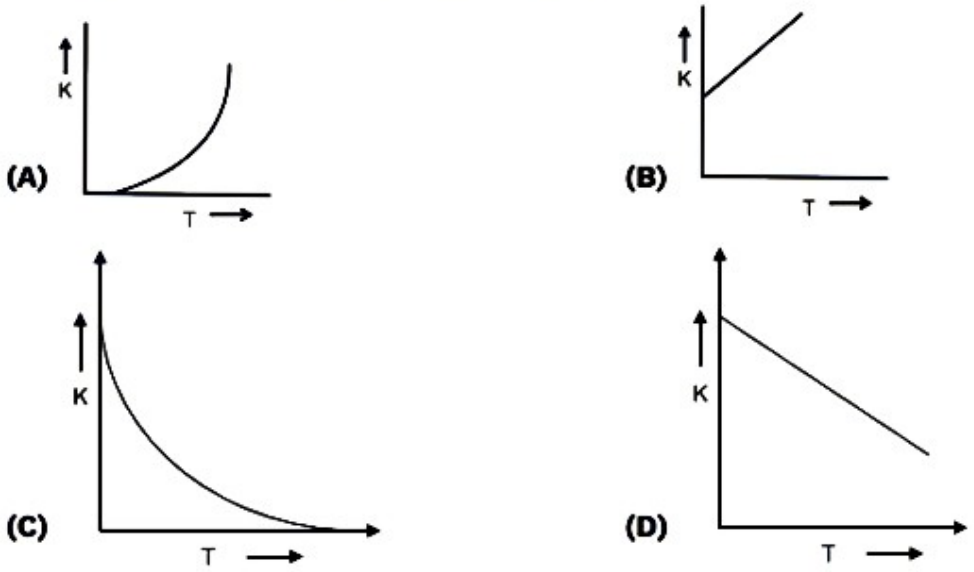


We know that, as temperature increases, Maxwell curve broadens.
only those particle convert into product.

whose Kinetic energy ≥ E_a of reaction

{area under the curve. ABCD}
= {Number of particles in activated state of temperature T_1}
and area of DAFG = Number of particles in activated states at temperature T_2.
Clearly (area of DEFG) > (Area of ABCD)
⇒ Number of particles in activated sate increases
Hence reaction rats increases.

Q43 **Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is [JEE 2010]**



Sol: **(A)**

As temperature(T) Increases => K increases (non-linearly) or exponentially according to Arrhenius equation $k = A.e^{\frac{E_a}{RT}}$

Q44 **The rate of a reaction doubles when its temperature changes from 300 K to 310 K. Activation energy of such a reaction will be [jee 2010]**

($R = 8.314 JK^{-1}mol^{-1}$ and $\log 2 = 0.301$)

(A) 66.5 kJ mol^{-1} **(B) 53.6 kJ mol^{-1}**

(C) 48.6 kJ mol^{-1} **(D) 58.5 kJ mol^{-1}**

Sol: **(B)**

K double

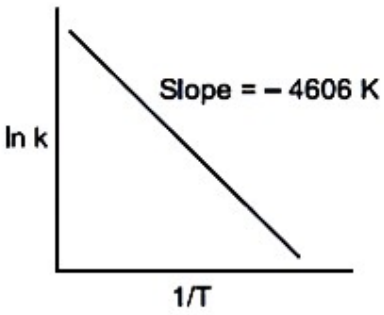
$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log\frac{2}{1} = \frac{E_a}{2.303 \times 8.314}\left(\frac{1}{300} - \frac{1}{310}\right)$$

$$E_a = \frac{0.3 \times 8.31 \times 300 \times 310 \times 2.303}{10 \times 1000} KJ/mol$$

$E_a = 53.6$ kJ mol^{-1}

Q45 **For a reaction, consider the plot of ln k versus (1/T) given in the figure. If the rate constant of this reaction at 400K is 10^{-5} s^{-1}, then the rate constant at 500 K is :** **[JEE Main Online 2019]**



(A) 10^{-4} s^{-1} **(B) 2×10^{-4} s^{-1}** **(C) 4×10^{-4} s^{-1}** **(D) 10^{-6} s^{-1}**

Sol: **(A)**

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$$

$$\text{Slope} = \left(-\frac{E_a}{R}\right) = -4606$$

$$\ln\frac{K_2}{K_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log\frac{k_2}{k_1} = \frac{4606}{2.303} \times \frac{100}{400 \times 500}$$

$$\log\frac{k_2}{k_1} = 1$$

$$\frac{k_2}{k_1} = 10$$

$$\Rightarrow k_2 = 10 \times k_1$$

$$k_2 = 10 \times 10^{-5}$$

$$k_2 = 10^{-4}\,s^{-1}$$

Temperature coefficient (m):

For a reaction.

$$\boxed{\mu = \frac{(k)_{T+10}}{(k)_T} \Rightarrow \mu > 1}$$

Generally $\boxed{2 \le \mu \le 3}$:

$(k)_{T+10} = \mu.\ (k)_T$

$\because$ Rate $\propto$ k

$\Rightarrow$ $(\text{Rate})_{T+10} = \mu\ .\ (\text{Rate})_T$

Points to remember

Means on every 10 kelvin increase in temperature the rate becomes μ times.
(if μ is temperature independently)

Q46 **The rate of a chemical reaction doubles for every 10°C rise of temperature. If the temperature is raised by 50°C, the rate of the reaction increases by about** **[AIEEE 2011]**

(A) 10 times **(B) 24 times** **(C) 32 times** **(D) 64 times**

Sol: **(C)**

As, $\mu = \frac{(k)_{T+10}}{(k)_T}$

then $50 = 5 \times 10$

$(\text{Rate})_{T+50} = 2 \times 2 \times 2 \times 2 \times 2\ (\text{Rate})_T = 32(\text{Rate})_T$

Catalyst :

The substance which increases the rate of a reaction without being consumed in the reaction is known as catalyst.

Inhibitor :

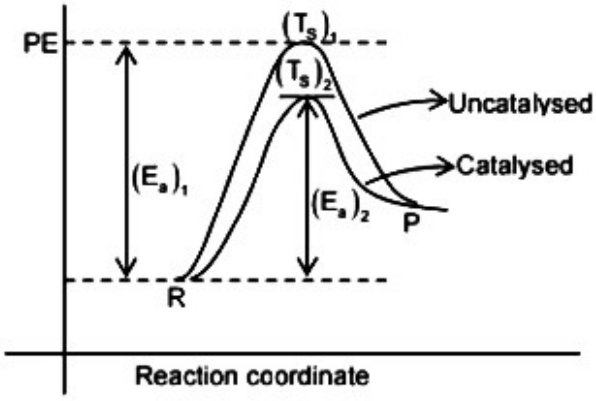
The substance which decreases the rate of reaction without being consumed in the reaction.

Effect of catalyst :

A catalyst provides an alternative path for the reaction of which E_{Th} is lesser than that of original path.

$\because \quad E_a = E_{Th} - E_R$

Means the activation energy of catalysed path is lesser than that of uncatalyzed path.



$\because \quad E_{a_2} < E_{a_1}$

$\because \quad k = A \cdot e^{-\frac{E_a}{RT}}$

$\Rightarrow \quad k_2 > k_1$

Mathematically :

Uncatalyzed : $\ln k_1 = \ln A - \frac{E_{a_1}}{RT}$

Catalyzed : $\ln k_2 = \ln A - \frac{E_{a_2}}{RT}$

$$\boxed{\ln \frac{k_2}{k_1} = \left(\frac{E_{a_1} - E_{a_2}}{RT} \right)}$$

$K_{net} = k\,[H^+]_{catalyst}$

Q47 **A catalyst lowers the activation energy for a certain reaction from 75 Kcal mol^{-1} to 15 Kcal mol^{-1}. What will be the effect on the rate of reaction at 27°C, other things being equal?**

Sol: $E_{a_1} = 75$ kcal/mol $E_{a_2} = 15$kal / mol

$$\ln\frac{k_2}{k_1} = \frac{E_{a_1} - E_{a_2}}{RT} = \frac{60 \times 10^3}{2 \times 300} = 100$$

$$\Rightarrow k_2 = k_1 \times e^{100}$$

$$\text{rate} \propto k \Rightarrow r_2 = r_1.e^{100}$$

Progress of a reversible reaction :

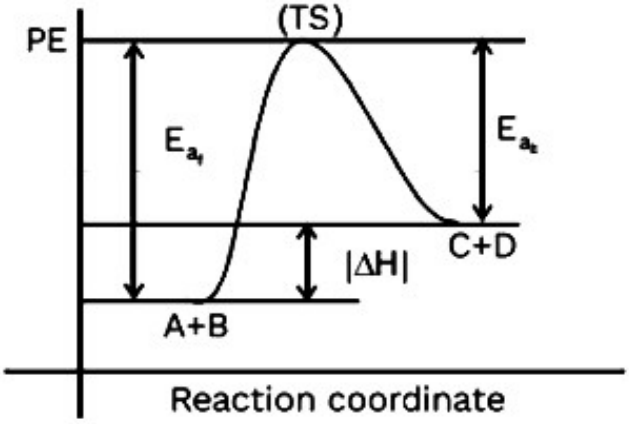
Example : $A + B \rightleftharpoons C + D : \Delta H > 0$

Forward reaction:

$$A + B \xrightarrow{E_{a_f}} C + D \; ; \; \Delta H > 0$$

Backward reaction:

$$C + D \xrightarrow{E_{a_b}} A + B \quad ; \quad \Delta H < 0$$



E_{af} = activation energy of forward direction

E_{ab} = activation energy of backward direction

$$\Rightarrow \boxed{\Delta H = \left(E_{a_f} - E_{a_b}\right)}$$

Progress of a complex reaction :

A complex reaction is actually combination of multiple sequential elementary reactions.

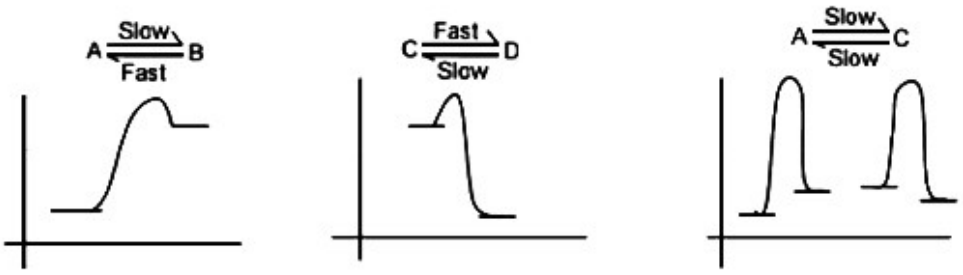
i.e., Number of mechanism steps = Number of peaks in graph.

If endothermic $\Rightarrow \Delta H > 0 \Rightarrow E_P > E_R$

If exothermic $\Rightarrow \Delta H < 0 \Rightarrow E_R > E_P$

Slow reactions have greater E_a and hence higher barrier height & vice-versa.

i.e.,



Q48 **For, R $\longrightarrow$ P ; $\Delta H < 0$**

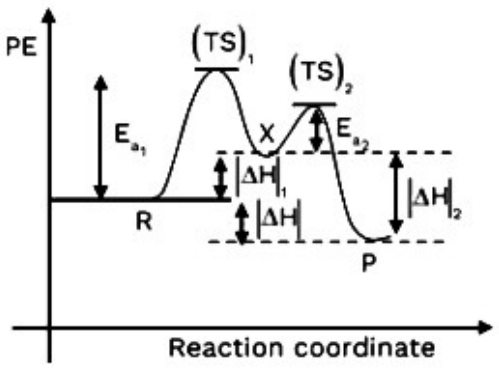
Mechanism :

(1) R $\xrightarrow{\text{Slow}}$ X ; $\Delta H > 0$

(2) X $\xrightarrow{\text{Fast}}$ P ; $\Delta H < 0$

Draw most suitable graph

Sol:



For R $\longrightarrow$ P : $\Delta H < 0$

Given reaction is exothermic

Hence $E_R > E_P$

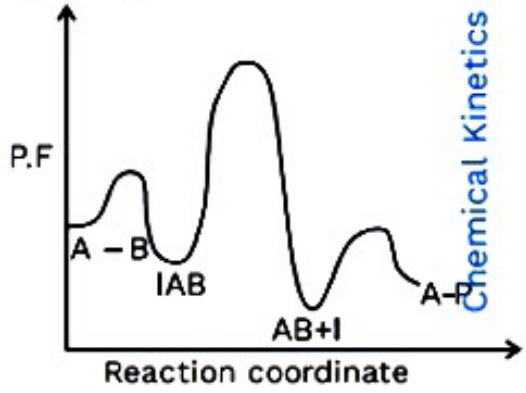
Since it is two step process, so there will be one intermediate and two transition state

Q49 **The following mechanism has been propose for the exothermic catalyzed complex reaction.**

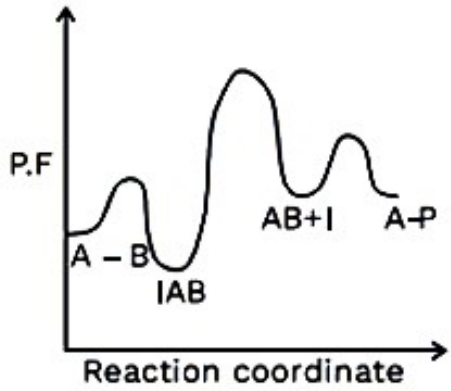
$$A + B \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} IAB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

If k_1 is much smaller than k_2. The most suitable qualitative plot of potential energy (P.E.) versus reaction coordinate for the above reaction.

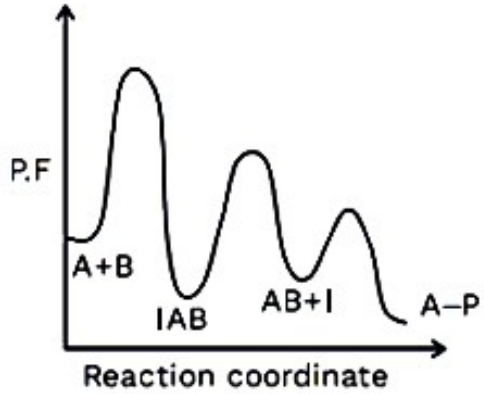
(A)



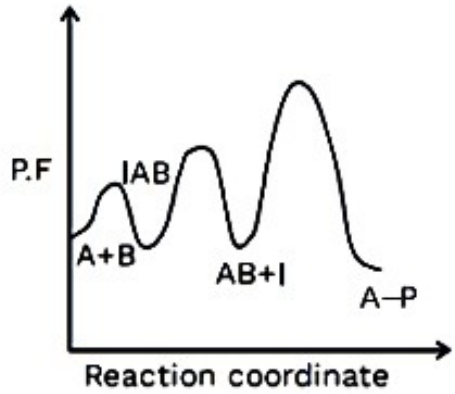
(B)



(C)



(D)



Sol: (A)

Overall reaction exothermic

$\Delta H < 0 \Rightarrow E_R > E_p$

$k_1 << k_2$ (given)

$\Rightarrow R_2 << R_3$ as $k = A \cdot e^{\frac{E_a}{RT}}$

Rate Determining Step method :

Used to determine rate law of a complex reaction when complete mechanism is given.

R.D.S → Rate Determining step (slowest step of mechanism)

$$k = A \cdot e^{-\frac{E_a}{RT}}$$

Step-1: Write down rate law of R.D.S. (elementary)

Step-2: It no intermediate is present in this rate law then it is final rate law, otherwise substitute the intermediate from other given equilibrium step.

Step-3: Repeat step (2) until all intermediates are substituted.

Q50 For $2NO + O_2 \longrightarrow 2NO_2$

Mechanism :

(i) $2NO \underset{k_2}{\overset{k_1}{\rightleftharpoons}} N_2O_2$ **; (fast eqm)**

(ii) $N_2O_2 + O_2 \xrightarrow{k_3} 2NO_2$ **; (slow)**

Sol: Step-(ii) $\Rightarrow$ RDS

So, Rate $= k_3[N_2O_2]^1 [O_2]^1$

$\downarrow$

intermediate

$\Rightarrow$ Not final rate law.

From step (i) :

$$k_{eq} = \frac{k_1}{k_2} = \frac{[N_2O_2]^1}{[NO]^2} \Rightarrow [N_2O_2] = \frac{k_1}{k_2}[NO]^2 \text{ put}$$

$$\text{Rate} = k_3\left(\frac{k_1}{k_2}[NO]^2\right)[O_2]^1$$

$$\text{Rate} = k_{net}[NO]^2[O_2]^1$$

$\because$ No intermediate $\Rightarrow$ final rate law

So, order $= 2 + 1 = 3$

$$\text{also, } k_{net} = \frac{k_1 \times k_3}{k_2}$$

$\because$ $k = A \times e^{\frac{E_a}{RT}}$

$$A_{net}\, e^{-\frac{E_{a_{net}}}{RT}} = \frac{\left(A_1.e^{-\frac{E_a}{RT}}\right)\left(A_3.e^{-\frac{E_{a_3}}{RT}}\right)}{\left(A_2.e^{-\frac{E_{a_2}}{RT}}\right)}$$

$$A_{net}\, e^{-\frac{E_{a_{net}}}{RT}} = \left(\frac{A_1.A_3}{A_2}\right).e^{-\frac{E_{a_1}+E_{a_3}-E_{a_2}}{RT}}$$

$$\text{So, } A_{net} = \frac{A_1 \times A_3}{A_2}$$

$$\text{also, } E_{a_{net}} = \left(E_{a_1} + E_{a_3} - E_{a_2}\right)$$

Radioactivity :

It is totally based upon observations.

Spontaneous disintegration (decay) of **unstable nuclei** by emission of radiation like α, β, γ etc.

This phenomenon is not affected by any external factor like temperature, pressure, physical size etc. so it was concluded that radioactivity is a nuclear phenomenon.

i.e. if ${}^{A}_{Z}X$ is radioactive then

$$\left.\begin{matrix} X_{(s)} \\ X_{(l)} \\ X_{(g)} \\ X^{+} \\ X^{-} \end{matrix}\right\} \Rightarrow \text{All will be radioactive}$$

Also, if a radioactive atom is present in a molecule, then the molecule will be radioactive too

Example :

${}^{12}_{6}C$ & ${}^{14}_{6}C$

Stable Radioactive

So, same compounds will be formed for both isotopes.

$$\text{So, } \left.\begin{matrix} {}^{14}CH_4 \\ {}^{14}CO_2 \\ {}^{14}CCl_4 \end{matrix}\right\} \Rightarrow \text{Radioactive}$$

Points to remember

Since E_a of elementary reactions are always positive so for an overall Complex reaction,

$E_{a_{net}}$ may be positive or negative or even zero.

Note: All radioactive decay obey 1st order kinetics but Arrhenius equation has no significance in radioactivity.

Let, ${}_{Z}X^{A}$ is a radioactive substance then,

$$X \longrightarrow \text{Products}$$

$t = 0$ N_o

$t = t$ N_t

where, $N_o \longrightarrow$ initial number of nuclei (atoms)

$N_t \longrightarrow$ Remaining number of nuclei at time t

$\Rightarrow$ Decayed nuclei till time $t = (N_o - N_t)$

$\because$ 1st order

Rate $\propto (N)^1$

$$\Rightarrow \left(-\frac{dN}{Dt}\right) = \lambda.N \Rightarrow -\int_{N_o}^{N_t} \frac{1}{N} dN = \lambda \int_{t=0}^{t=t} dt$$

$$\text{So, } \lambda.t = \ln \frac{N_o}{N_t}$$

$N_t = N_o \, e^{-\lambda.t}$

$\lambda.t = 2.303 \log \frac{N_o}{N_t}$

Where, $\lambda \rightarrow$ decay constant (time^{-1})

Note : Unlike rate constant (k), decay constant is inde

Points to remember

So, $lt = \begin{cases} \ln \frac{N_o}{N_t} \\ \ln \frac{n_o}{n_t} \\ \ln \frac{A_o}{A_t} \\ \ln \frac{W_o}{W_t} \end{cases}$

Half-life of radioactive substance $\left(t_{\frac{1}{2}} \text{ or } t_{50\%} \right)$

i.e., $N_t = N_o \left(1 - \frac{1}{2}\right)$

put $\boxed{t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}}$ Here half life is independent of initial concentration

Also, $t_{\frac{3}{4}} = 2\frac{t_1}{2}$

Also, after x half lives

$\boxed{N_t = \frac{N_o}{2^x}}$

Points to remember

Every t_x is independent of initial number of nuclei.

Mean life/Average life ($t_{avg.}$) :

$t = 0 \quad \underset{1 \text{ mole}}{X} \xrightarrow{t_{\frac{1}{2}} = 10 \text{min}}$

at t = 10 min, X = (1–0.5) = 0.5 mole at t = 10 min total 3.01×10^{23} atoms decayed.
Hence life of particle was different hence average life of particle.

$\boxed{t_{avg} = \frac{\text{Sum of lives of all atoms}}{\text{Number of atoms}}}$ and $\boxed{t_{avg} = \frac{1}{\lambda}}$

Activity of radioactive substance (A):
It is defined as the rate of disintegration.

$A = \left(-\frac{dN}{dt}\right)$

i.e. $A_o = \lambda.N_o = \lambda.(n_o \times N_A) = \lambda.\left(\frac{\omega_o}{M} \times N_A\right) \Rightarrow$

$A_t = \lambda.N_t = \lambda.(n_t \times N_A) = \lambda.\left(\frac{\omega_t}{M} \times N_A\right)$

Where,

$n \rightarrow$ moles

$\omega \rightarrow$ mass (gm)

M → atomic mass

Units of activity :

dps → disintegration per sec.

dpm→ disintegration per minute

Ci → Curie (1 Ci = 3.7×10^{10} dps)

Bq → Bequerel(1Bq =1 dps)

$A = \lambda . N$

Note: in numerical, generally dps or dpm is used.

Specific activity:

Activity of one gm radioactive substance.

i.e. $A = \lambda . \left(\frac{W}{M} \times N_A\right)$

For specific activity, w = 1 gm

Q51 **The half-life of Tc^{99} is 6.0hr . The delivery of a sample of Tc^{99} from the reactor to the nuclear medicine lab of a certain hospital takes 3.0 hr. What is the minimum amount of Tc^{99} that must be shipped for the lab to receive 10.0 mg ?**

(A) 20.0 mg **(B) 15.0 mg** **(C) 14.1 mg** **(D) 12.5 mg**

Sol: **(C)**

Reactor $\xrightarrow{3hr.}$ Lab

$W_o = ?$ $W_t = 10$ mg

So, $\frac{\ln 2}{6} \times 3 = \ln \frac{\omega_o}{10}$

$\ln \frac{\omega_o}{10} = \ln \sqrt{2}$

$\omega_0 = 10\sqrt{2}$

$= 14.1$ mg

Q52 **The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 S. The mass of an atom of this radioisotope is 10^{-25}kg. The mass (in mg) of the radioactive sample is :**

[JEE-2011]

Sol: $A = 10^{10}$dps

$t_{avg} = 10^9 \text{sec} \Rightarrow \lambda = 10^{-9}\ \text{sec}^{-1}$

$$A = \lambda \times \left(\frac{\omega}{M} \times N_A\right) = \lambda \times \left(\frac{\omega_N}{M / N_A}\right)$$

$$10^{10} = 10^{-9}\left(\frac{\omega}{10^{-25} \times 10^3}\right)$$

$\omega = 10^{-3}\text{gm} = 1\text{mg}$

Applications of radioactivity:

(1) Carbon dating

Used to determine age of once living objects like fossils, dead bodies, wooden articles etc

Age → The time between object's death and at present.

Q53 **An old wood sample was found to contain C–14 activity of x dps of present while a freshly cut tree sample has C- 14 activity of Y dps. If t $\frac{1}{2}$ of C–14 is T years then find the age of old wood sample.**

Sol: For old sample

$$\lambda . T = \ln\frac{A_o}{A_t}$$

A_o = Activity of old at the time of death

A_t = activity of old at present.

Assumption:

In every living object, the C–14 content is same & is constant from last 50,000 years.

So, activity of old just before death (A_0) = Activity of any living $\Rightarrow \frac{\ln 2}{T} \times t = \ln\frac{y}{x}$

Q54 **Wooden article and freshly cut tree show activity of 7.6 and 15.2 min^{-1} gm^{-1} of carbon ($t_{1/2}$ = 5760 years) respectively. The age of article in years, is :**

(A) 5760

(B) $5760 \times \left(\frac{15.2}{7.6}\right)$

(C) $5760 \times \left(\frac{7.6}{15.2}\right)$

(D) $5760 \times (15.2 - 7.6)$

Sol: **(A)**

$t_{1/2}$= 5760 years

$$kt = \ln\left(\frac{\omega_0}{\omega_t}\right)$$

$$\text{so, } \frac{\ln 2}{t_{1/2}} \times t = \ln\frac{\omega_0}{\omega_t}$$

$$\frac{\ln 2}{5760} \times t = \ln\frac{15.2}{7.6}$$

t = 5760 years

(2) Uranium dating :

Used to determine age of non-living object's like stone, rocks, ores, minerals etc.

Age :

The time period between object's formation (t = 0) & at present (t = t)

Note :

In a nuclear reaction (spontaneous & artificial both). Law of mass conservation is not valid but in such reactions, atomic number (z) & mass number (A) remain Conserved.

i.e. $a.{}^{A_1}_{Z_1}P + b.{}^{A_2}_{Z_2}Q \longrightarrow c.{}^{A_3}_{Z_3}R + d{}^{A_4}_{Z_4}S$

So, $a(Z_1) + b(Z_2) = c.(Z_3) + d.(Z_4)$

& $a(A_1) + b.(A_2) = c(A_3) + d(A_4)$

Some important particles :

Atomic number	Mass number	Particle
1	1	Proton (P)
0	1	Neutron (n)
– 1	0	Beta (β)
2	4	Alpha (∝)
1	2	Deuteron (D)
1	0	Positron (β^+)
0	0	Gamma radiation

Q55 **Identify the emitted particle in following nuclear reaction :**

a) $_{92}^{238}U \longrightarrow {}_{90}^{234}Th + {}_{Z}^{A}X$

b) $_{6}^{14}U \longrightarrow {}_{7}^{14}N + {}_{Z}^{A}X$

Sol: a) On balancing atomic number, we get
$1(92) = 1(90) + 1(Z) \Rightarrow Z = 2$
On balancing mass number, we get
$1(238) = 1(234) + 1(A) \Rightarrow A = 4$
i.e. $X \Rightarrow \propto$– particle

b) On balancing atomic number, we get
So, $1(6) = 1(7) + 1(Z) \Rightarrow Z = (-1)$
On balancing mass number, we get
$1(14) = 1(14) + 1(A) \Rightarrow A = 0$
i.e. $X \Rightarrow \beta$ particle

Q56 **The total number of α and β particles emitted in the nuclear reaction** $^{238}_{92}U \longrightarrow ^{214}_{82}Pb$. **[JEE 2009]**

$$^{238}_{92}U \longrightarrow ^{214}_{82}Pb + x^{4}_{4}\alpha + y^{0}_{-1}\beta$$

Sol: On balancing atomic number, we get

$1(92) = 1(82) + x(2) + y(-1)$

$\Rightarrow 2x - y = 10$

On balancing mass number, we get

$1(238) = 1(214) + x(4) + y(0)$

On solving both equation, we get

$\Rightarrow$ Number of α particles $(x) = 6$,

$\Rightarrow$ Number of β particles $(y) = 2$

(3) Artificial radioactivity

Not necessarily follow 1st order kinetics.

A target nucleus is bombarded by a specific particle due to which it dissociates into other nuclie emitting some radiations.

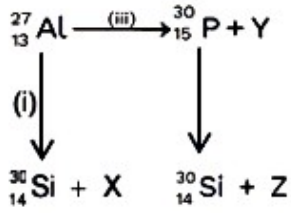
i.e., $A + x \longrightarrow (B + C ...) + y$

Other representation :

$$A \xrightarrow{(x,y)} (B + C ...)$$

Or $A(x, y)B$

Q57 **Bombardment of aluminum by α -particle leads to its artificial disintegration in two way (i) and (ii) as shown. Products X, Y, and Z respectively are,** **[JEE 2011]**



(A) proton, neutron, positron **(B) neutron, positron, proton**

(C) proton, positron, neutron, **(D) positron, proton, neutron**

Sol: **(A)**

$$^{27}_{13}Al + ^{4}_{2}\alpha \longrightarrow ^{30}_{14}Si + ^{A_1}_{Z_1}X$$

$Z_1 = 1$

$A_1 = 1 \quad \Rightarrow \quad X = \text{Proton}$

$$^{27}_{13}Al + ^{4}_{2}\alpha \longrightarrow ^{30}_{15}P + ^{A_2}_{Z_2}Y$$

$Z_2 = 0$

$A_2 = 1$

$\Rightarrow$ Y = Neutron

$$^{30}_{15}P \longrightarrow ^{30}_{14}Si + ^{A_1}_{Z_1}Z$$

$Z_1 = 1$

$A_1 = 0$

$\Rightarrow$ Z = positron

Q58 **The number of neutrons accompanying the formation of $_{54}Xe^{139}$ and $_{38}Sr^{24}$ from the absorption of slow neutron by $_{92}U^{235}$ followed by nuclear fision is**

[JEE 1999]

(A) 0 **(B) 2** **(C) 1** **(D) 3**

Sol: **(D)**

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{139}_{54}Xe + ^{94}_{38}Sr + x.\left(^{1}_{0}n\right)$$

On balancing mass number, we get

So $1(235) + 1(1) = 1(139) + 1(94) \times (1)$

$\Rightarrow x = 3$

Types of radioactive decay :

(1) α -decay :

It is shown by very heavy nuclei only (Z > 83)

In general,

$$^{A}_{Z}X \longrightarrow ^{4}_{2}\alpha + ^{A-4}_{Z-2}Y$$

i.e.,Z decreases by 2 & A decreases by 4

Example :

$$^{234}_{92}U \xrightarrow{-\alpha} ^{230}_{90}Th$$

Also, in each α-emission, 2p ↓ & 2n↓

So, $\left(\frac{n}{P}\right)\uparrow$

(2) β-decay :

A beta particle is exactly similar to electron but is emitted from the nucleus.

Whenever a neutron converts into a proton within the nucleus, a beta particle is emitted.

i.e., $^{1}_{0}n \longrightarrow ^{1}_{1}P + {}^{0}_{-1}\beta$

In general ;

$$^{A}_{Z}X \longrightarrow ^{0}_{-1}\beta + ^{A}_{z+1}Y$$

i.e.,Z increase by 1 but mass no. is same.

Example :

$$^{14}_{6}C \xrightarrow{-\beta} ^{14}_{7}N$$

Unstable stable

Also, In each β. emission, n↓ & 'P ↑

So, $\left(\frac{n}{P}\right)\downarrow$

(3) γ-decay :

γ-radiations are just EMR of very high energy.

Whenever a nucleus emits or aborts a particle, if goes into a relatively lower stable state and to regain stability, it emits energy In the form of γ-radiations.

Example :

$$^{238}_{92}U \xrightarrow{-\alpha} ^{234}_{90}Th^{*} \xrightarrow{\rightarrow} ^{234}_{90}Th + \gamma$$

Lesser stable more stable

Here Th & Th* are nuclear isomers of each other

(4) Positron decay :

A positron is antiparticle of electron

$(m)_{\beta^+} = 9.1 \times 10^{-31}$ kg

$(q)_{\beta^+} = +1.6 \times 10^{-19}$C

Whenever a proton converts into a neutron within the nucleus, a positron is emitted.

i.e. ${}_1^1p \longrightarrow {}_0^1n + {}_1^0\beta^+$

In general :

$${}_Z^AX \longrightarrow {}_1^0\beta^+ + {}_{Z-1}^AY$$

i.e z decreases by 1 but mass no same

Eg. ${}_{12}^{23}Mg \xrightarrow{-\beta^+} {}_{11}^{23}Na$

Unstable Stable

also, In each positron emission, P↓ & n↑

So, $\left(\frac{n}{P}\right)\uparrow$